



COSMETIC PRODUCT SAFETY REPORT

For:

DERMOTIN BIOROID

In accordance with Regulation (EC) 1223/2009 of the European Parliament and of the Council of 30 November 2009 on cosmetic products

Responsible Person: Republic of Serbia: Clicky doo, Zvezdarskih Jelki 3/2, Beograd

EU: Clicky EOOD, Traditsa 6, Sofia, Bulgaria

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Safety Assessor: Vesna Milanković

Part A. Cosmetic Product Safety Information

Trade name: DERMOTIN BIOROID

Volume: 50ml

Product category:

Manufacturer: BIO-TEO centar d.o.o., Mitropolita Mraovića 1a, 11000 Belgrade, Serbia

1) Qualitative and quantitative composition of the product

INCI	Concentration of raw material in the product (%)	Raw material composition (%)	CAS	EINECS/ELINCS	Function	Impurities
Aqua	75.29	100	7732-18-5	231-791-2	Water phase	N/A
Aqua	10	30-60	7732-18-5	231-791-2	Disolver	N/A
Alcohol		30-60	64-17-5	200-578-6	Disolver	
Achillea Millefolium Extract		10-15	84082-83-7	282-030-6	Active	
Aesculus Hippocastanum Seed Extract		7-10	8053-39-2	232-497-7	Active	
Calendula Officinalis Flower Extract		4-8	84776-23-8	283-949-5	Active	
Capsella Bursa-Pastoris Extract		4-8	84604-19-3	283-297-1	Active	
Geranium Robertianum Extract		3-5	84650-11-3	283-492-1	Active	
Chamomilla Recutita Flower Extract		3-5	84082-60-0	282-006-5	Active	
Stearic Acid	2.16	100	57-11-4	200-313-4	Oil phase	N/A
Glycerin	2	100	56-81-5	200-289-5	Humctant	N/A
Cetyl Alcohol	1.6	100	36653-82-4	253-149-0	Oil phase	N/A
Glyceryl Stearate SE	1.4	100	11099-07-3	234-325-6	Emuslifier	N/A
Synthetic Beeswax	1.2	100	71243-51-1	275-286-5	Oil phase	N/A
Panthenol	0.4	100	81-13-0 / 16485-10-2	201-327-3 / 240-540-6	Active	N/A
Phenoxyethanol	1	88.5-91.5	122-99-6	204-589-7	Preservative	N/A
Ethylhexylglycerin		8.5-11.5	70445-33-9	408-080-2	Stabilizer	

Prunus Amygdalus Dulcis Oil	1		8007-69-0 / 90320-37-9	- / 291-063-5	Oil phase	N/A
Urea	0.4		57-13-6	200-315-5	Active	N/A
Triethanolamine	0.88		102-71-6	203-049-8	pH adjuster	N/A
Carbomer	0.36		9007-20-9 / 9003-01-4 / 76050-42-5 / 9062-04-8 / 9007-16-3 / 9007-17-4	- / - / - / - / -	Rheology modifier	Ethyl Acetate 0.23%, Cyclohexane 0.02%
Potassium Sorbate	0.3		24634-61-5 / 590-00-1	246-376-1 / -	Preservative	Formaldehyde 0.1%
Menthol	0.15		1490-04-6 / 2216-51-5 / 89-78-1 / 15356-70-4	216-074-4 / 218-690-9 / 201-939-0 / 239-388-3	Active	N/A
Retinyl Palmitate	0.1		79-81-2	201-228-5	Active	N/A
Tocopheryl Acetate	0.5		7695-91-2 / 58-95-7	231-710-0 / 200-405-4	Antioxidant	N/A
Pogostemon Cablin Leaf Oil	0.08		8014-09-3 / 84238-39-1	- / 282-493-4	Active	N/A
Lavandula Hybrida Oil	0.08		8022-15-9 / 91722-69-9	- / 294-470-6	Active	N/A
Melaleuca Alternifolia Leaf Oil	0.05		85085-48-9 / 8022-72-8 / 68647-73-4	285-377-1 / - / -	Active	Limonene 2%
Eugenia Caryophyllus Leaf Oil	0.05		8015-97-2 / 8000-34-8 / 84961-50-2	- / - / 284-638-7	Active	Eugenol 83.29%
Sodium Gluconate	0.5		527-07-1 / 14906-97-9	208-407-7 / 238-976-7	Stabilizer	N/A
Tocopherol	0.5		0	200-201-5 / 240-747-1 / 233-466-0 / 204-299-0 / 215-798-8 / - / 218-197-9 / 200-412-2 / -	Antioxidant	N/A

*Limonene and Eugenol must be labeled

2) Physical and chemical characteristics and stability of the cosmetic product

a) Physical and chemical characteristics of the raw materials

For all raw materials used for production of this cosmetic product technical documentation (SDS, TDS, CoA, etc.) provided by the supplier of cosmetic ingredients is available and is attached to this document. After evaluating the documents provided, it was concluded that all the ingredients used for production of cosmetic product DERMOTIN BIOROID met the requirements of quality and purity criteria, as well as that they contain the information needed for an adequate assessment of the safety of the cosmetic product.

INCI	Chemical name	Molecular formula	Molecular weight	Water solubility	Vapour pressure (Pa)	log Pow
Aqua	Water	H ₂ O	18.02 g/mol	N/A	N/A	N/A
Aqua	Water	H ₂ O	18.02 g/mol	N/A	N/A	N/A
Alcohol	ethanol	C ₂ H ₆ O	46.07	789 000 mg/L, completely miscible in water	5 726 Pa	-0.35
Achillea Millefolium Extract	N/A	N/A	N/A	soluble	N/A	N/A
Aesculus Hippocastanum Seed Extract	N/A	N/A	N/A	N/A	N/A	N/A
Calendula Officinalis Flower Extract	0	0	N/A	soluble, insoluble	59.3 mm Hg (ethanol extr)	N/A
Capsella Bursa-Pastoris Extract	N/A	N/A	N/A	N/A	N/A	N/A
Geranium Robertianum Extract	N/A	N/A	N/A	soluble	N/A	N/A
Chamomilla Recutita Flower Extract	N/A	N/A	N/A	soluble	N/A	N/A
Stearic Acid	Stearic acid	C ₁₈ H ₃₆ O ₂	0	< 0.05 mg/L	< 8.54E-06	8.23
Glycerin	Glycerol	C ₃ H ₈ O ₃	92.09	Completely miscible with water at 25 °C. The compound is also completely miscible with methanol, ethanol, and the isomers of propanol, butanol, and pentanol.	N/A	log POW = -1.75 at 25 °C
Cetyl Alcohol	Hexadecan-1-ol	CH ₃ (CH ₂) ₁₅ OH	242.44	<1.0 mg/L at 23 °C, 0.024 mg/L at 25 °C (literature)	N/A	6.7; 7.3 at 23 °C
Glyceryl Stearate SE	1,3-dihydroxypropan-2-yl octadecanoate; 2,3-	C ₂₁ H ₄₂ O ₄	358.56	< 1 mg/L	< 0.0001 Pa at 20 °C	6.1

	dihydroxypropyl octadecanoate					
Synthetic Beeswax	n/a	n/a	N/A	Insoluble	N/A	N/A
Panthenol	2,4-dihydroxy-N-(3-hydroxypropyl)-3,3-dimethylbutanamide	C ₉ H ₁₉ NO ₄	205.25	509 g/L	5.6 E-05 Pa at 25 °C	-1.06
Phenoxyethanol	2-Phenoxyethanol	C ₈ H ₁₀ O ₂	138.17	26.7 g/l at 20 °C	N/A	log Pow = 1.16
Ethylhexylglycerin	1,2-Propanediol, 3-(2-ethylhexyloxy)	C ₁₁ H ₂₄ O ₃	204.31	Limited solubility in water (cca. 0.1%); highly soluble in organic solvents, such as alcohols, glycols, and glycol ethers.	N/A	log POW = 2.4
Prunus Amygdalus Dulcis Oil	N/A	N/A	N/A	Insoluble	N/A	N/A
Urea	urea	CH ₄ N ₂ O	60,06 g/mol	624 000 mg/L	1.2 x 10 ⁻⁵ mmHg (0.0016 Pa) at 25 °C.	-1.73
Triethanolamine	2-[bis(2-hydroxyethyl)amino]ethan-1-ol	C ₆ H ₁₅ NO ₃	149.2 g/mol	>1000 g/l at 20 °C	< 0.0003 hPa at 21 °C	-2.3
Carbomer	Polyacrylic acid	N/A	7×10 ⁵ to 4×10 ⁹ Da	soluble	N/A	N/A
Potassium Sorbate	potassium (2E,4E)-hexa-2,4-dienoate	C ₆ H ₇ O ₂ K	150.22 g/mol	1.95 g/L	0 hPa	-1.72
Menthol	2-isopropyl-5-methylcyclohexanol	C ₁₀ H ₂₀ O	0	0.397 g/L, moderately soluble	19 Pa at 25 °C	Log Kow 3.15 at 25°C
Retinyl Palmitate	(2E,4E,6E,8E)-3,7-dimethyl-9-(2,6,6-trimethylcyclohex-1-en-1-yl)nona-2,4,6,8-tetraen-1-yl hexadecanoate	C ₃₆ H ₆₀ O ₂	N/A	The substance appears insoluble in water. Results for water solubility <1mg/l, out of experimental data and for calculated data = 2.8E-11mg/l at 25°C.	extrapolated vapour pressure 5.1E-7 hPa at 20°C	15.51
Tocopheryl Acetate	3,4-Dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-benzopyran-6-yl acetate	C ₃₁ H ₅₂ O ₃	472.7	Insoluble in water	N/A	log POW = 12 (estimated)
Pogostemon Cablin Leaf Oil	N/A	N/A	N/A	insoluble	7.91 Pa	6.6
Lavandula Hybrida Oil	N/A	N/A	N/A	2.012 mg/L	358 Pa	4.8
Melaleuca Alternifolia Leaf Oil	N/A	N/A	N/A	Insoluble	N/A	N/A

Eugenia Caryophyllus Leaf Oil	N/A	N/A	N/A	insoluble	N/A	N/A
Sodium Gluconate	N/A	C6H11NaO7	218.14	soluble	N/A	minus 5.99 (estimated)
Tocopherol	(2R)-2,5,7,8-tetramethyl-2-[(4R,8R)-4,8,12-trimethyltridecyl]-3,4-dihydro-2H-1-benzopyran-6-ol; (2R)-2,5,8-trimethyl-2-[(4R,8R)-4,8,12-trimethyltridecyl]-3,4-dihydro-2H-1-benzopyran-6-ol; (2R)-2,7,8-trimethyl-2-[(4R,8R)-4,8,12-trimethyltridecyl]-3,4-dihydro-2H-1-benzopyran-6-ol; (2R)-2,8-dimethyl-2-[(4R,8R)-4,8,12-trimethyltridecyl]-3,4-dihydro-2H-1-benzopyran-6-ol	N/A	430.7 Da	1.3 µg/L	1.80E-8 hPa at 25°C	12.2

b) Physical and chemical characteristics of the cosmetic product

Parameter	Value
Appearance	Cream
Colour	White to beige
Odour	Pleasant
pH	6.78
Density	N/A
Viscosity	N/A

c) Cosmetic product's stability

A laboratory study of stability and compatibility was performed, and the study protocol and results are attached to this document.

Stability study protocol description:

- Accelerated test, in original packaging, for 12 weeks, on +5C, room temperature (25C) and +40C,.
- Centrifugation measurement
- Measurement of organoleptic parameters (color, odor), pH, viscosity and phase separation were measured periodically
- Microbiology testing at the beginning and at the end of the test period

Based on the results obtained, it is concluded that the product DERMOTIN BIOROID is stable and compatible with the packaging under reasonably applicable conditions of use and storage on 5C, 25C and 40C.

Due to slight change in color after 12 weeks on 40C shelf life should not be longer than 24 months.

3) Microbiological quality

a) Microbiological quality of raw materials

Raw materials used for DERMOTIN BIOROID are followed with technical documentation (SDS, TDS, CoA, etc.) provided by the supplier which is available and attached to this document. The technical documentation was reviewed, based on which it was concluded that all the ingredients of the cosmetic product represent a satisfactory microbiological quality. The production of the cosmetic product is carried out in accordance with the principles of good manufacturing practice, which excludes the possibility of microbiological contamination during the production process.

b) Microbiological quality of the cosmetic product

Quantitative and qualitative limits are based on the European Standard EN ISO 17516:2014 Cosmetics – Microbiology – Microbiological limits. The European Standard EN ISO 17516:2014 was approved by CEN on 9 August 2014 and at present is widely used 149 by the cosmetics industry as international standard as given (SCCS Notes of Guidance 12th revision):

Types of microorganism	Products specifically intended for children under three years of age, the eye area or the mucous membranes	Other products
Total Aerobic Mesophilic Microorganisms (Bacteria plus yeast and mould)	$\leq 1 \times 10^2$ CFU per g or ml ^a	$\leq 1 \times 10^3$ CFU per g or ml ^b
<i>Escherichia coli</i>	Absence in 1 g or 1 ml	Absence in 1 g or 1 ml
<i>Pseudomonas aeruginosa</i>	Absence in 1 g or 1 ml	Absence in 1 g or 1 ml
<i>Staphylococcus aureus</i>	Absence in 1 g or 1 ml	Absence in 1 g or 1 ml
<i>Candida albicans</i>	Absence in 1 g or 1 ml	Absence in 1 g or 1 ml
Due to inherent variability of the plate count method, according to USP Chapter 61 or EP Chapter 2.6.12, Interpretation of results, results considered out of limit if a > 200 CFU/g or ml, b > 2 000 CFU/g or ml. NOTE When colonies of bacteria are detected on Sabouraud Dextrose agar, Sabouraud Dextrose agar containing antibiotics may be used.		

Microbiological testing of the cosmetic product DERMOTIN BIOROID was conducted. The results are proving the compliance of the product with the limits named above for the category „Other products“. The laboratory results report is attached to this document.

Challenge test: The efficacy of preservation system used for cosmetic product DERMOTIN BIOROID was conducted using method ISO 11930:2019 - Cosmetics - Microbiology - Evaluation of the antimicrobial protection of a cosmetic product. The result obtained confirmed the efficacy of the antimicrobial protection of a cosmetic product DERMOTIN BIOROID. The laboratory results are attached to this document.

4) Impurities, traces, information about the packaging material

a) Impurities and traces in the raw materials

Raw materials used for the manufacturing of the cosmetic product DERMOTIN BIOROID are well-known chemical substances and mixtures that have long been used as raw materials in the cosmetic industry. Based on the technical documentation (SDS, TDS, CoA, etc.) provided by the suppliers of used raw materials, the presence of impurities and traces was determined for which the Margin of Safety (MoS) was calculated using the Threshold of Toxicological Concern (TTC) calculation:

INCI	Cramer Class	NOAEL	SED	MoS
Ethyl Acetate	I	0.03	0.000165724	181.0236526
Cyclohexane	I	0.03	2.88216E-05	1040.886002
Formaldehyde	III	0.000025	3.6027E-06	6.939240014

The Threshold of Toxicological Concern (TTC) calculation is a scientifically acceptable method for assessing the risk of systemic toxic effects that may arise from the use of chemical substances present in very small quantities. This evaluation method is considered acceptable in case of lack of adequate toxicological data necessary to assess the risk of intentionally using additional cosmetic ingredients as well as the impurities present in them.

The Cramer class was determined by using software TOXTREE.

Results obtained for the MoS indicate safe exposure to ingredients with identified impurities.

b) Heavy metals content

Heavy metals content of the cosmetic product DERMOTIN BIOROID was determined for Pb, Cd, As, Cr, Hg, and Ni. The results are given as less than 0.5mg/kg for each component which is considered acceptable. The laboratory results report is attached to this document.

Cosmetic product DERMOTIN BIOROID is produced in Republic of Serbia where heavy metals content determination is mandatory. Results obtained are acceptable according to the limits set by the cosmetic regulation in Republic of Serbia.

c) The relevant characteristics of packaging material

Cosmetic product DERMOTIN BIOROID was packed in a white plastic cosmetic container with a lid made from the same material. The container is being placed in a cardboard box. Elements used for packaging described are well known and are being widely used as packaging for pharmaceutical and cosmetic products for a long time.

The packaging supplier submitted the technical documentation for each element that is attached to this document and is describing the used materials.

5) Normal and reasonably foreseeable use

Cosmetic product DERMOTIN BIOROID is intended to be used as cosmetic cream for the skin around anal area without subsequent washing after application.

Instructions for use written on the cosmetic product:

„Apply twice a day, along with regular hygiene. The cream is of herbal origin and can be used for a long period of time without interruption.

Warnings and precautions written on the cosmetic product:

„Hypersensitivity to any of the ingredients. The product is intended for external use only. Wash hands after use. Avoid product contact with eyes. If contact occurs, wash with plenty of water. In case of extended irritation, seek medical help. “

The cosmetic product DERMOTIN BIOROID is presented in a way that clearly communicates its purpose. Precise and clearly understandable instructions are given, based on which it is considered that it cannot be misused.

6) Exposure to the cosmetic product

Duration of product application	Leave-on
Body part intended for application	Skin around anal area
Reasonably foreseeable application on other body parts	N/A
Duration and frequency of application	2 times a day
Relative daily exposure (mg/kg bw/day)*	40.3
Target population	Adults
Retention factor	1

*estimated according to the SCCS Notes of Guidance for the testing of cosmetic ingredients and their safety evaluation (12th revision)

7) Exposure to the cosmetic products ingredients

Systemic Exposure Dose (SED) was calculated using the formula:

$$\text{SED (mg/kg bw/day)} = A \text{ (mg/kg bw/day)} * C \text{ (\%)/100} * DA \text{ (\%)/100}$$

Where parameters are:

A (mg/kg bw/day) – Relative daily exposure

C (%) – Maximum concentration of the ingredient in the final product

DA (%) – Dermal absorption

Dermal absorption (DA) and no observed adverse effect level (NOAEL) values were determined by detailed analysis of the toxicological profiles of substances that are given in the part 8. If oral bioavailability and dermal absorption values were not known, the default value of 50% was used.

Margin of safety (MoS) was calculated using the formula:

$$\text{MoS} = \text{NOAEL} / \text{SED}.$$

INCI naziv sastojka	Maksimalna koncentracija sastojka u proizvodu (%)	SED (mg/kg bw/day)	NOAEL (mg/kg bw/day)	MoS
Aqua	81.29	30.138587	N/A	N/A
Alcohol	6	2.4018	1200	499.625281
Achillea Millefolium Extract	1.5	0.60045	400	222.055
Aesculus Hippocastanum Seed Extract	1	0.20015	30	149.8875843
Calendula Officinalis Flower Extract	0.8	0.32024	200	624.5316013
Capsella Bursa-Pastoris Extract	0.8	0.32024	N/A	N/A
Geranium Robertianum Extract	0.5	0.20015	200	333.08
Chamomilla Recutita Flower Extract	0.5	0.20015	50	249.8126405
Stearic Acid	2.16	0.864648	500	192.75
Glycerin	2	0.8006	100	124.9063203
Cetyl Alcohol	1.6	0.64048	2000	1040.886002
Glyceryl Stearate SE	1.4	0.56042	500	297.3960006
Synthetic Beeswax	1.2	0.48036	400	832.7088017
Panthenol	0.4	0.16012	1000	2081.77
Phenoxyethanol	0.915	0.3662745	N/A	N/A
Ethylhexylglycerin	0.115	0.0460345	25	543.0709577
Prunus Amygdalus Dulcis Oil	1	0.4003	N/A	N/A
Urea	0.4	0.16012	1125	7025.980515
Triethanolamine	0.88	0.352264	50	141.9390003
Carbomer	0.36	0.144108	N/A	N/A
Potassium Sorbate	0.3	0.12009	N/A	N/A
Menthol	0.15	0.060045	375	6245.316013
Retinyl Palmitate	0.1	0.008006	1.43	178.616038
Tocopheryl Acetate	0.5	0.20015	180	299.7751686
Pogostemon Cablin Leaf Oil	0.08	0.032024	810	8431.176618
Lavandula Hybrida Oil	0.08	0.032024	160	1665.417603
Melaleuca Alternifolia Leaf Oil	0.05	0.020015	20	333.0835207
Eugenia Caryophyllus Leaf Oil	0.05	0.020015	2.5	124.9063203
Sodium Gluconate	0.5	0.20015	500	2498.126405
Tocopherol	0.5	0.20015	250	1249.063203

According to the guidelines of the World Health Organization and the Scientific Committee on Consumer Safety (SCCS), a Margin of Safety (MoS) value of at least 100 is considered a sufficient indicator of safe use of the substance.

8) Toxicological profile of the substances

INCI	ALCOHOL
Presence in the annexes	0
Description	According to the International Cosmetic Ingredient Dictionary and Handbook (Gottschalck and McEwen 2004), Alcohol is the name given by convention to ethanol as used in cosmetics, and Alcohol Denat. is the name given to denatured Alcohol for use in cosmetics. Alcohol Denat. is prepared by adding one or more denaturing agents to potable Alcohol, making it unfit for beverage or internal human medicinal use. Alcohol may be completely denatured (CD Alcohol) or specially denatured (SD Alcohol) depending on the type(s) of denaturant(s) used.
Oral toxicokinetics	Following any route of intake resulting in an elevated blood ethanol level . Inhalation: In a study to determine the fractional uptake of ethanol during inhalation exposure, human volunteers were exposed to ethanol vapour concentrations of 10 -12mg/l (6000ppm) for periods of 1 hour. Experiments were done with and without a prior dose of ethanol by the oral route. The fractional absorption was typically around 55% (variation less than 10% within and between individuals) and was found to be not detectably affected by variations in tidal volume, breathing frequency nor the existing blood ethanol concentration. The concentration of ethanol in expired air did not fall below 30%.
Inhalation toxicokinetics	In a study to determine the fractional uptake of ethanol during inhalation exposure, human volunteers were exposed to ethanol vapour concentrations of 10 -12mg/l (6000ppm) for periods of 1 hour. Experiments were done with and without a prior dose of ethanol by the oral route. The fractional absorption was typically around 55% (variation less than 10% within and between individuals) and was found to be not detectably affected by variations in tidal volume, breathing frequency nor the existing blood ethanol concentration. The concentration of ethanol in expired air did not fall below 30%.
Dermal absorption	In a study to assess the skin penetration potential of ethanol, an in vitro study was carried out using excised pig skin and radiolabelled ethanol. Ethanol penetration was greater under occlusion conditions than non-occlusive conditions, as might be expected. Absorption rates were around 21% and 1% of applied doses respectively.
Skin irritation	not irritating
Eye irritation	not irritating
Skin sensitisation	not sensitising
Acute toxicity	In a guideline acute oral toxicity study, the LD50 was determined to be 10470 mg/kgbw when dosed as a 95% solution in water. Dermal: LD50 17 100 mg/kg bw. Inhalation: In an acute inhalation study that approximated to guideline, ethanol vapour was found to be of very low acute toxicity to male and female rats. The LC50

	(4hr) was established to be around the 117 -125mg/l and the LC0 around 62mg/l. It is worthy of note that the LD50 is well above the lower explosive limit (LEL).
Repeated dose studies	Oral treatment of rats for 14 weeks with 0, 5, 10 or 20 mg/Kg of a mixture containing 16.25% USP ethanol, The NOAEL for the study is determined at 10 ml/Kg. Most data available on ethanol is via the oral route of exposure. Much is at high doses which limits its value to risk assessment of ethanol as a chemical substance. From the data available, it is possible to surmise that ethanol is of repeat dose low toxicity by the oral route, with a lowest reported NOAEL in animals of 2400 mg/kg for rats.
Mutagenicity/genotoxicity	OECD Guideline 471 (Bacterial Reverse Mutation Assay): negative. In vivo mammalian germ cell study: cytogenicity / chromosome aberration: Ethanol is unlikely to be a dominant lethal mutagen, at least up to the maximum tolerated dose.
Carcinogenicity	Rats: No effect level >3000mg/kg. Mice (B6C3F1): Females: NOAEL>44000mg/kg (cancer), Males: NOAEL>4250mg/kg (based on historic control data), Males BMDL10=1400mg/kg (based on concurrent control data). Overall, there is no significant evidence to warrant a classification of ethanol for cancer in the context of the relevant classification and labelling regulations for chemical substances.
Reproductive toxicity	Overall, ethanol in drinking water at concentrations up to 15% (equivalent to 20.7 g/kg/day) had no demonstrable effect on fertility in this two-generation study. Groups of female Sprague-Dawley rats were given liquid diets with 15, 25, or 36% ethanol-derived calories (E15, E25, and E36 groups, respectively), or without ethanol (pair-fed isocaloric (PF15, PF25, PF36) or ad libitum (C) control) for 3 weeks prior to mating, and throughout 21 days of gestation. Prenatal ethanol exposure affected fetal skeletal ossification at exposure levels lower than those required to affect fetal body weight and length, although the significance of these changes for long-term bone health is unknown. No statistically significant effects on fetal growth or ossification were seen in the fetuses of dams ingesting about 5.2 g ethanol/kg bw/day from 3 weeks prior to mating to GD 21.
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. ECHA https://echa.europa.eu/sl/registration-dossier/-/registered-dossier/16105/7/10/2 2. OECD SIDS, Ethanol, Assessment Report https://hvpchemicals.oecd.org/ui/handler.axd?id=2602cc56-d998-4e67-bd78-5454ef3f8f9a
POD (NOAEL)	2400mg/kg bw/day

INCI	GLYCERIN
Presence in the annexes	0
Description	Glycerol is widely distributed in food as a natural constituent, and it has undergone review and approval for use as both a direct and indirect food additive and is GRAS by US FDA. Animal studies involving single or repeated exposure by various routes indicated a low order of toxicity, the possible target sites (particularly following repeated oral administration) being the kidney and gastrointestinal tract. Glycerol appears to be of generally low oral toxicity in humans.
Oral toxicokinetics	Glycerin is rapidly absorbed in the intestine and the stomach, distributed throughout the extracellular fluids through much of the body. It is mostly

	metabolized by the liver and kidneys with the remainder excreted in urine. Free glycerin is naturally present in humans, primarily in plasma.
Inhalation toxicokinetics	The NOAEC in rats for local and systemic effects after 13-week exposure is 662 mg/m ³ .
Dermal absorption	0.1
Skin irritation	Not dermally irritating to rabbits when applied at concentrations up to 100% to up to 30% of the body surface. Mild dermal irritant at 100% in guinea pigs.
Eye irritation	Undiluted glycerin was not irritating when administered to the eyes of human subjects. There was a strong burning and stinging sensation, with tear production but no injury was observed.
Skin sensitisation	A moisturizer containing 65.9% glycerin was not sensitizing to human subjects. Natural and synthetic glycerin were not sensitising to white male guinea pigs at 0.1%.
Acute toxicity	The reported oral LD50 of glycerin ranged from 2530-58400 mg/kg in rats, 4090-38000 mg/kg in mice, 27000 mg/kg in rabbits, and 77500 mg/kg in guinea pigs. The dermal LD50 of glycerin in rats was reported to be > 21900 mg/kg and >18700 mg/kg in rabbits. The approximate Lt50 for rats was determined to be 423 min for exposure to glycerin vapors at 11.0 mg/L. The intraperitoneal LD50 of glycerin in rats ranged from 4420-10100 mg/kg and 8600-9500 mg/kg in mice. The subcutaneous LD50 of glycerin was 100 mg/kg in rats and ranged from 91-10000 mg/kg in mice. The intravenous LD50 of glycerin ranged from 5200-6600 mg/kg in rats, 4250-6700 mg/kg in mice, and 53000 mg/kg in rabbits.
Repeated dose studies	In the best available dietary study, groups of 22 rats (Long-Evans)/sex/treatment received 5, 10 and 20% glycerol (natural or synthetic) in their diet (males 2000, 4000 and 8000 mg/kg bw; females 2500, 5000 and 10000 mg/kg bw) for 2 years. Although the results were not described in detail, based on this limited dietary study it can be concluded that no adverse effects were observed at up to 10,000 mg/kg bw. Dermal: The effects following repeated dermal application of glycerin was examined. There were no effects noted in rabbits dosed 8 hours/day, 5 days/week for 45 weeks with dose levels as high as 4.0 ml/kg. Using a density of 1.2611 g/cm ³ at 20 °C, a dose of 4.0 ml/kg corresponds to 5040 mg/kg/day.
Mutagenicity/genotoxicity	Glycerin was not genotoxic in multiple Ames tests using multiple strains of <i>S. typhimurium</i> at concentrations up to 50 mg/plate. It was not genotoxic in a cytogenetic assay, X-linked HGPRT, sister chromatid exchange assay, unscheduled DNA synthesis assay, and chromosome aberration test at concentrations up to 1.0 mg/mL.
Carcinogenicity	Glycerin administered in the feed of rats at doses up to 20% in feed for 1 year or up to 10 g/kg for 2 years did not increase the incidence of tumors. Orally administered glycerin, in concentrations up to 5%, had a potentiating effect on the carcinogenicity of 4NQO in mice.
Reproductive toxicity	No convincing evidence of reproductive effects have been seen in rats treated orally or dermally, or in mice fed glycerol during pregnancy. Glycerin is transferred across the placenta in small amounts. May cause some adverse reproductive effects based on animal data, but there was no evidence of teratogenicity. No effects on fertility and reproductive performance were observed in a two generation study with glycerol administered by gavage (NOAEL 2000 mg/kg bw/day).
Photosensitivity	N/A
Endocrine effects	N/A
Reference	(1) CIR Expert Panel. Safety assessment of glycerin as used in cosmetics - final report (January 14, 2015). (2) ECHA https://echa.europa.eu/sl/registration-dossier/-/registered-dossier/14481/7/6/1

POD (NOAEL)	reproduction 2000, oral 10000, dermal 5040
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INCI	POTASSIUM SORBATE
Presence in the annexes	V/4
Description	Potassium sorbate (E 202) is authorised for use as a food additives in the European Union. It is present in Annex V of 1223/2009 Directive with max. Allowed concentrations in ready for use preparations of 0.6% (acid).
Oral toxicokinetics	Sorbic Acid as well as Potassium Sorbate is rapidly and completely metabolized in the animal. The extrapolation of the metabolism of sorbic acid to potassium sorbate is considered not to be restricted in anyway, since the determination of potential toxicity is on the "sorbate" anion.
Inhalation toxicokinetics	N/A
Dermal absorption	N/A
Skin irritation	not irritating
Eye irritation	irritating
Skin sensitisation	not sensitising
Acute toxicity	Sorbic Acid and Potassium Sorbate show a very low toxicity in rats and mice in acute oral toxicity studies. The median lethal dose (LD50) of the Sorbic Acid is 10.50±1.96 g/kg body weight in the rat, while the LD50 of the Potassium salt has been determined at 3.2-10.5 g/kg body weight in the rat.
Repeated dose studies	EFSA Panel established a group ADI expressed as 11 mg sorbic acid/kg bw per day for sorbic acid (E 200) and its potassium salt (E 202).
Mutagenicity/genotoxicity	Sorbic Acid and its salts are non-mutagenic and non-clastogenic in vitro and in vivo.
Carcinogenicity	Potassium sorbate and sorbic acid are not carcinogenic.
Reproductive toxicity	An EOGRTS (extended one generation reproductive toxicity study) compliant with GLP was performed in CD/Cl:CD rats following OECD TG 443 regarding cohorts 1A and 1B (reproductive toxicity). In all generations, neither test item-related mortality nor effect on clinical signs in the animals and their pups were observed.
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. EFSA Panel "Opinion on the follow-up of the re-evaluation of sorbic acid (E200) and potassium sorbate (E202) as food additives" https://efsa.onlinelibrary.wiley.com/doi/epdf/10.2903/j.efsa.2019.5625 2. ECHA https://echa.europa.eu/registration-dossier/-/registered-dossier/11008/7/9/1
POD (NOAEL)	11mg/kg bw/day

INCI	Stearic Acid
Presence in the annexes	None
Description	Stearic acid C18. Fatty acids are typically unbranched, long chain organic acids with different chain length. They are found in all living organism fulfilling three fundamental roles. Besides their function as part of molecules like phospholipids and glycolipids important for the cell-structure, they are often precursors of signalling molecules such as prostanooids in animals or phytohormones in plants. The third and best understood role of fatty acid is their role as an energy source, particularly in higher animals and plants.
Oral toxicokinetics	Taken together the experimental and calculated data show that fatty acids are almost completely absorbed after oral intake, whereas only limited dermal uptake has to be expected.

Inhalation toxicokinetics	LC50 >0.1521 mg/L
Dermal absorption	poorly absorbed through skin, with a measured rate of less than 1 % after 24-hours exposure
Skin irritation	not irritating
Eye irritation	not irritating
Skin sensitisation	not sensitising
Acute toxicity	LD50 >6000 mg/kg bw
Repeated dose studies	Combined repeated dose and reproductive/developmental toxicity screening test, NOAEL 1000mg&kg bw&daz, duration unknown
Mutagenicity/genotoxicity	All available data on genotoxicity showed that fatty acids are not genotoxic.
Carcinogenicity	N/A
Reproductive toxicity	Combined repeated dose and reproductive/developmental toxicity screening test, NOAEL 1000mg&kg bw&daz, duration unknown
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. ECHA https://echa.europa.eu/registration-dossier/-/registered-dossier/15163/7/7/1
POD (NOAEL)	1000mg/kg bw/day, rats, oral, duration unknown

INCI	Glyceryl Stearate SE
Presence in the annexes	None
Description	Monoglyceryl monoester.
Oral toxicokinetics	No data on oral absorption
Inhalation toxicokinetics	N/A
Dermal absorption	Monoglyceryl monoesters can act as penetration enhancers.
Skin irritation	not irritating
Eye irritation	Slight to moderate effects on the conjunctivae (redness and chemosis) and slight effects on the iris (congestion) were observed in all animals 1 h after test substance instillation. In a single animal, effects on the iris persisted up to 48 h and reversed thereafter. No effects on cornea were observed in any animal tested. Not irritating
Skin sensitisation	not sensitising
Acute toxicity	> 5 000 mg/kg bw
Repeated dose studies	NOAEL, 90 days, rat> corresponding to doses of 5835 and 5725 mg/kg bw/day in males and females, respectively, as calculated from the reported food consumption and body weights
Mutagenicity/genotoxicity	In vitro> Genotoxicity: negative. Cytotoxicity> Not cytotoxic
Carcinogenicity	There is no evidence that the members of the Glyceride category induce gene mutations in bacteria or chromosome aberrations in mammalian cells, as the results of all genotoxicity studies were consistently negative. Furthermore, in the available repeated dose toxicity studies and developmental studies, no substance-related increases in the incidence of hyperplasia or pre-neoplastic lesions were observed. Justification for selection of carcinogenicity via oral route endpoint: No carcinogenicity study is required, since the substance is not mutagenic and no hyperplasia or pre-neoplastic lesions were observed in any of the available studies. The available and relevant studies do not indicate that the members of the Glyceride category fulfil the criteria for classification as germ cell mutagen or that they are able to induce hyperplasia and/or pre-neoplastic lesions.
Reproductive toxicity	NOAEL, parental fertility 1 000 mg/kg bw/day (actual dose received). Embryotoxic / teratogenic effects:no effects

Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. ECHA https://echa.europa.eu/sl/registration-dossier/-/registered-dossier/2133/7/2/2 2. CIR Exper Panel, Amended Safety Assessment of Monoglyceryl Monoesters as Used in Cosmetics
POD (NOAEL)	1000 from reproduction toxicity , duration unknown

INCI	Cetyl Alcohol
Presence in the annexes	0
Description	It is a 16-carbon alcohol, known also as 1 -hexadecanol and n-hexadecyl alcohol. Cetyl Alcohol is used as an emollient to prevent drying and chapping of the skin because of its water-binding property. Because Cetyl and Stearyl Alcohols have structural similarities, the literature review may be applicable to Cetyl Alcohol. Long-chain aliphatic alcohols (C ₁₂ and C ₁₆) induced minimal ocular and skin irritation but no sensitization or comedogenicity in rabbits; no mutagenic effects were noted in the Ames assay. In a subchronic percutaneous toxicity study, a product formulation (C ₁₂ -alcohol content) induced erythema and mild desquamation. Clinical studies of long-chain alcohols (C ₁₂ and C ₁₆) indicated a low order of skin irritation and sensitization. Also, results were negative in clinical phototoxicity and photosensitization studies of products containing these alcohols.
Oral toxicokinetics	No bioaccumulation potential based on study results. Following skin application of lauryl alcohol about 2.84 % of the administered dose was absorbed. Of this absorbed dose >90% was excreted in expired air (CO ₂). A similar trend was observed with the other alcohols tested, including hexadecan-1-ol. Absorption decreased with increasing carbon chain length and was affected by solvent and concentration. Good absorption: Approximately 15% of the alcohol was unchanged during its passage through the mucosal cells of the small intestine;
Inhalation toxicokinetics	N/A
Dermal absorption	Of a dose of 5% 1-14C-cetyl alcohol applied to the skin of nude mice for 24 hours under occlusive conditions, 1% was absorbed.
Skin irritation	Following a 4 hour semi-occlusive exposure of Kalcol 6098 to rabbit skin there was no evidence of skin irritation between 24 and 72 hours after patch removal. Kalcol 6098 is not a skin irritant according to EU or GHs criteria.
Eye irritation	Kalcol 6098 is not an eye irritant according to EU or GHS criteria. The result is a read across from hexadecanol (CAS 36653-82-4).
Skin sensitisation	In a reliable study performed using a protocol similar to OECD guideline 406, Dobanol 25 was not sensitising to the skin of guinea pigs after a topical challenge with 2.5% when tested using the method of Magnusson and Kligman. (1977)
Acute toxicity	The rat oral LD50 for Kalcol 6098 is >2000 mg/kg. At this dose level there were no signs of toxicity. The rat 1 hour inhalational LC50 for tetradecan-1-ol is >1.5 mg/l. There were no signs of toxicity and findings at gross necropsy were unremarkable. The rabbit dermal LD50 (24 hour occluded) for Alfol 14 was approx. 8000 mg/kg.

Repeated dose studies	For hexadecan-1-ol, oral NOAELs were > 4257 and > 4567 mg/kg bw/day in male and female rats respectively in a 90-day repeated dose toxicity test in which a somewhat limited range of endpoints was evaluated (Scientific Associates 1966a). the systemic No-Observed-Adverse-Effect-Level following dermal administration of fatty alcohol blend for a minimum of 90 days was considered to be 1000 mg/kg/day (the highest dose tested).
Mutagenicity/genotoxicity	In a reliable study, performed according to OECD guideline 471, the C16 alcohol Kahlcol 6098 did not increase the reverse mutation rate in histidine dependent bacterial strains of Salmonella typhimurium in the presence or absence of metabolic activation at concentrations up to 5000 µg/plate. This concentration was not cytotoxic. Fatty alcohol blend has been tested according to OECD 474 and under GLP. Male and female mice were dosed with 500, 1000 and 2000 mg/kg bw. No increase in the number of micronucleated PCE was observed (1000 PCE scored per animal). It is concluded that the test substance is negative for the induction of micronuclei under the conditions of the test. No toxicity to bone marrow or general toxicity was observed.
Carcinogenicity	the weight of evidence across the category suggests that hexadecan-1-ol is not carcinogenic. This is based on a large set of various types of repeated dose studies across the category which do not offer any evidence of treatment-related induction of hyperplasia / pre-neoplastic lesions for hexadecan-1-ol or structurally related alcohols (though reporting is limited in many cases). In addition, hexadecan-1-ol does not have any genotoxic effects.
Reproductive toxicity	In a reliable study conducted to the draft OECD guideline 422, a parental NOAEL of > 2000 mg/kg bw/day (highest dose tested) was determined for male and female rats. No adverse effects were observed on reproductive parameters and the NOAEL for reproductive and developmental effects was also > 2000 mg/kg bw/day. In a reliable study, development was assessed as part of a combined repeat dose and reproductive/developmental toxicity study, conducted according to draft OECD guideline 422. The NOAEL for maternal and foetotoxicity in rats was 2000 mg/kg bw/day (highest dose level). There was no evidence of teratogenicity from the limited examination of the pups that was carried out. The result is read across from octadecan-1-ol (CAS 112-92-5).
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. Publication: Percutaneous absorption of aliphatic compounds, Iwata Y, Moriya Y, Kobayashi T, Cosmet. Toiletries 102(2): 53-68 (1987). 2. ECHA https://echa.europa.eu/sl/registration-dossier/-/registered-dossier/15942/1/1 3. CIR Expert Panel, Final Report on the Safety Assessment of Cetearyl Alcohol, Cetyl Alcohol, Isostearyl Alcohol, Myristyl Alcohol, and Behenyl Alcohol
POD (NOAEL)	2000 from Reproduction toxicity as worst case, oral high, duration unknown

INCI	Synthetic Beeswax
Presence in the annexes	0
Description	Beeswax is a GRAS food ingredient. In cosmetics, it is used in baby products, bath and fragrance preparations, eye and facial makeup, hair, skin, nail, shaving and suntan preparations, at concentrations of less than 0.1% to greater than 50%. A total content of 27 to 40% monoesters, 9 to 23

	%hydroxymonoesters, 7 to 16% diesters, 3.9 % hydroxydiesters, 11 to 28% hydrocarbons, 1 to 18% saturated, unbranched free fatty acids, 4 to 8% of other substances and < 0.3% free fatty alcohols have been reported in beeswaxes from different bees species. Experimental biochemical and toxicological studies on beeswax as such are lacking. However, the Panel considered that the safety of beeswax could be evaluated on the basis of published toxicological studies on its main constituents. An EFSA Committee considered that the data on beeswax itself were insufficient to establish an ADI, but concluded that the safety of beeswax could be assessed, based on available scientific literature on the main constituents of beeswax and plant waxes showing chemical structural similarities to beeswax. They could not reach a conclusion about the potential allergenicity of beeswax as the available information was very limited. However, the committee concluded that the use of beeswax as an additive for the existing food uses is not of safety concern. (EFSA 2007).
Oral toxicokinetics	Oral: low.
Inhalation toxicokinetics	N/A
Dermal absorption	N/A
Skin irritation	mild acute skin irritation
Eye irritation	Mild irritation
Skin sensitisation	N/A
Acute toxicity	Acute oral doses of 5 g/kg Beeswax produced no toxicity in rats.
Repeated dose studies	A cleansing cream formulation containing 13% Beeswax was used in a sixweek dermal toxicity study using 20 rats. A 0.3 g/kg dose of the test material was applied to the shaved skin of each of 10 rats, once per day, five days per week, ASSESSMENT: CANDELILLA WAX, CARNAUBA WAX, JAPAN WAX, AND BEESWAX 29 for a total of 30 applications. Ten other rats were used as untreated controls. Observations were made throughout the test period for general appearance, behavior, and toxicologic signs. Body weights were obtained weekly and blood samples were taken before the end of the study. All animals survived the study, and had normal appearance and behavior. External appearance of the skin at the site of application did not differ from the skin of untreated controls. Gross and microscopic examinations detected no abnormalities, and it was concluded that the formulation produced no systemic toxic effects.
Mutagenicity/genotoxicity	Beeswax was not mutagenic in tests using Salmonella typhimurium or Saccharomyces cerevisiae with and without metabolic activation.(g3
Carcinogenicity	Threads impregnated with Beeswax (controls) or 3-methylcholanthrene (MC) plus Beeswax were implanted in the cervixes of six-week-old virgin female C341HeJ mice. Epithelialization of the cervical canal occurred after two and four weeks with both Beeswax and Beeswax plus MC. Invasive carcinoma developed in mice treated with Beeswax plus MC after four weeks of treatment, but not in the control (Beeswax-treated) animals.(91) In a similar experiment, Beeswax and Beeswax plus MC-impregnated thread were implanted into the cervixes of mice. In Beeswax-treated mice, and invasive squamous carcinoma was found in one of the 35 mice tested after four and six weeks; MC plus Beeswax induced 58% carcinoma of the cervix in treated mice.”)
Reproductive toxicity	N/A
Photosensitivity	A mascara formulation containing 10.0% Beeswax was tested for photosensitivity on 68 panelists. Applications of 0.1 ml/cm ² were made for 24

	h, evaluated for irritation, and then irradiated with a 150 W quartz bulb lamp with a continuous emission of 300-370 nm at a distance of 12 in from the test site. Scoring of the irradiated sites were made after 48 h, and the process was repeated six times. After a 10-day nontreatment period, a challenge application was applied to an adjacent, untreated site for 24 h. This new site was then irradiated. After 72 h, none of the 68 subjects developed photosensitivity.(104
Endocrine effects	N/A
Reference	1. CIR Expert Panel, Final Report on the Safety Assessment of Candelilla Wax, Carnauba Wax, Japan Wax, and Beeswax 2. https://efsa.onlinelibrary.wiley.com/doi/pdf/10.2903/j.efsa.2007.615
POD (NOAEL)	300mg/kg bw/day from 6weeks repeated dose study for a 13% beeswax cream. NOAEL for beeswax could be taken as 800 as worst case (it is 13% in the cream)

INCI	Phenoxyethanol
Presence in the annexes	V/29
Description	European Cosmetics Regulation Annex V (Line 29) lists phenoxyethanol as an approved preservative in the EU at concentrations up to 1%. In terms of Regulation (EC) No 1272/2008 (CLP) the following hazard classes are assigned to this ingredient (EU harmonized classification): Acute Tox. 4, Eye Irrit. 2.
Oral toxicokinetics	Tests with rats show that more than 75% and up to 99% of the phenoxyethanol after either oral or dermal exposure can be found unchanged in the urine together with small quantities of two substances to which the phenoxyethanol has metabolized. One of the metabolism products is phenoxyacetic acid.
Inhalation toxicokinetics	N/A
Dermal absorption	Rapidly absorbed through the rat skin.
Skin irritation	Not skin irritating (humans); slightly irritating (animals)
Eye irritation	Irritating (rabbit)
Skin sensitisation	Not sensitizing (human, animal)
Acute toxicity	Oral LD50 (rat): 1386 - 4013 mg/kg; Dermal LD50 (rat): 14300 mg/kg
Repeated dose studies	Dermal exposure is the usual exposure route for most cosmetic ingredients such as 2-phenoxyethanol and more relevant than oral exposure, which is often used as a surrogate exposure when animal safety studies are conducted with cosmetic ingredients. This view on the limited relevance of oral RDT data and dermal exposure as the relevant exposure route is supported by the toxicokinetic data (section 3.3.9), where an extensive first-pass metabolism after oral exposure of rats was found so that only traces of the haematotoxic agent 2-phenoxyethanol reached the systemic circulation. In contrast, after dermal exposure of rats even to low doses of 2-phenoxyethanol, up to 20% of the dose appeared in the blood as the parent compound (Kim et al. 2015, AR 9). Therefore, the 90-day dermal toxicity study in rabbits SCCS/1575/16 Final version of the Opinion on Phenoxyethanol with an adjusted NOAEL of 357 mg/kg bw/day is used as the key study for the MoS calculation.
Mutagenicity/genotoxicity	Nonmutagenic in the Ames test and in the mouse micronucleus test.
Carcinogenicity	Not expected to be carcinogenic.
Reproductive toxicity	Phenoxyethanol has shown damaging impacts on reproduction and developmental toxicity in animal tests with mice. In several reproduction studies with mice the impacts were decreasing body weight on the mice and their progeny as well as increased liver weight at high doses of between 1875 and 4000 mg/kg/day.

Photosensitivity	In clinical studies phenoxyethanol was nonphototoxic.
Endocrine effects	N/A
Reference	(1) Danish Ministry of Environment, EPA. A survey and health assessment of cosmetic products for children. Survey of Chemical Substances in Consumer Products, No. 88 2007. (2) CIR Expert Panel. Final Report on the Safety Assessment of Phenoxyethanol. JACT 9(2):259-277, 1990.
POD (NOAEL)	357 dermal

INCI	Ethylhexylglycerin
Presence in the annexes	0
Description	The safety of ethylhexylglycerin has been assessed by the CIR Expert Panel. The CIR Expert Panel evaluated the scientific data and concluded that this ingredient is safe as used in cosmetics at concentrations up to 8%. Although ethylhexylglycerin is generally considered safe, it has been found to be an irritant to the skin in two studies, causing allergic contact dermatitis.
Oral toxicokinetics	In an acute toxicokinetic study in vivo, the mean absorption of ethylhexylglycerin through the skin of rabbits was 0.02% at approximately 2-hour postapplication, and there were no signs of skin irritation. The amount of ethoxyglycerin in the plasma was below the detection limit at the end of the 4-hour application period.
Inhalation toxicokinetics	N/A
Dermal absorption	Over a range of 3 concentrations (44.65%, 47.15%, and 54.94%) applied to human skin in vitro, mean penetration rates of 2.38, 8.19, and 20.38 µg/cm ² /h were reported. Rabbit The mean absorption through skin in this group was 0.02% at approximately 2 hours after the beginning of exposure.
Skin irritation	Non-irritating in HRIPT (up to 1%); mildly to non-irritating (undiluted, rabbit).
Eye irritation	Undiluted ethylhexylglycerin was severely irritating, but 5% ethylhexylglycerin was mildly irritating, to the eyes of rabbits.
Skin sensitisation	Not sensitizing (up to 50%, guinea pig and local lymph node assay); not sensitizing (0.4-1%, HRIPT). Positive patch test results were reported for dermatitis patients patch tested with ethylhexylglycerin at concentrations up to 10%. However, results were negative for control group.
Acute toxicity	No mortalities or exposure-related toxicological findings were observed in rats dosed orally with undiluted ethylhexylglycerin. Oral LD50 (rat) > 2000 mg/kg
Repeated dose studies	In rats dosed orally with ethylhexylglycerin at doses up to 1500 mg/kg for 28 days, the 100 mg/kg dose was defined as the NOAEL. Ethylhexylglycerin administered orally to rats, at doses up to 800 mg/kg/day, in a 13-week study, a NOAEL of 50 mg/kg/day (lowest dose) was derived.
Mutagenicity/genotoxicity	Not genotoxic in Ames tests using multiple strains of <i>S. typhimurium</i> and in the mouse lymphoma assay in vitro. Nonclastogenic in the micronucleus assay in vivo.
Carcinogenicity	N/A
Reproductive toxicity	In a preliminary test, the results of visceral and skeletal examinations in litters of female rats given oral doses of ethylhexylglycerin up to 800 mg/kg/day were negative. The NOAEL for developmental toxicity was considered to be 800 mg/kg/day. In the 1-generation developmental toxicity study in rats, NOEL was 50 mg/kg/day.
Photosensitivity	Not phototoxic or photoallergic (guinea pig).

Endocrine effects	N/A
Reference	(1) CIR Expert Panel. Safety assessment of alkyl glyceryl ethers as used in cosmetics. IJT 32(Suppl. 3):5-21, 2013.
POD (NOAEL)	50, 13 weeks

INCI	Prunus Amygdalus Dulcis Oil
Presence in the annexes	None
Description	Sweet Almond Oil is the fixed oil obtained from the ripe seed kernels of various species of Prunus. Almond kernels contain as much as 50% oil, which can either be physically expressed (35% yield) or chemically extracted (50% yield). Sweet Almond Oil consists mostly of triglycerides of oleic and linoleic acids. Oils derived from edible vegetables, fruits, seeds, tree, and ground nuts have been safely consumed by, and applied to the skin of, humans for thousands of years. The group of ingredients characterized as fats and oils are the glyceryl esters of fatty acids (triglycerides) normally found in plants. The raw oil may include diglycerides, monoglycerides, free fatty acids, plant sterols, pigments, glucosides, proteins, natural antioxidants, vitamins, and impurities. exposure to the oils from food use would result in a much larger systemic dose than that resulting from use in cosmetic products. Cocos nucifera(coconut) oil, oil from the dried coconut fruit, is composed of 90% saturated triglycerides. The primary constituents of coconut oil are trimyristin, trilaurin, tripalmitin, tristearin, and various other triglycerides. 11 About 90% of the triglycerides are saturated.
Oral toxicokinetics	Sweet Almond Oil was well-digested (97%) when compared to other oils. In another study, Sweet Almond Oil or whole almonds were added to a lipid-free basal diet at concentrations of 10% and 18%, respectively, and fed to groups of three rats for three days. Absorption of the Almond Oil was 98%-99% complete.
Inhalation toxicokinetics	N/A
Dermal absorption	The percutaneous absorption of Sweet Almond Oil was studied in adult albino rats. A solution containing this ingredient and physostigmine was applied to the skin of each animal; latency time to muscle contraction was then measured. When compared to other oil mixtures tested, Sweet Almond Oil-physostigmine had a relatively long latency time, indicating lower percutaneous absorption.
Skin irritation	Undiluted Sweet Almond Oil was tested for irritancy in groups of six male albino rabbits. The test material was applied under occlusion to the clipped intact and abraded dorsal skin of each animal. Twenty-three hours later, patches were removed; sites were scored at 24 and 48 hours. The Primary Irritation Indices (PIIs) for seven test samples of Sweet Almond Oil ranged from 0 to 0.18 (maximum score = 8), indicating that this ingredient is practically nonirritating to skin.'
Eye irritation	slightly irritating
Skin sensitisation	not sensitising
Acute toxicity	Undiluted Sweet Almond Oil and products containing 2%-18% Sweet Almond Oil were tested for acute oral toxicity. Results of these tests indicate that undiluted Sweet Almond Oil and products containing up to 18% Sweet Almond Oil are practically nontoxic
Repeated dose studies	N/A

Mutagenicity/genotoxicity	Bacterial Reverse Mutation Assay of Almond Sweet Extract (CAS 90320-37-9) - test item does not show any mutagenic nor pro-mutagenic activity.
Carcinogenicity	N/A
Reproductive toxicity	N/A
Photosensitivity	Results of photo patch tests of formulations containing up to 2% Sweet Almond Oil in 764 subjects indicated that these products do not manifest photosensitization.
Endocrine effects	N/A
Reference	1. CIR Expert Panel Safety Assessment of Prunus Amygdalus Dulcis (Sweet Almond) Meal Seed 2. ECHA https://echa.europa.eu/registration-dossier/-/registered-dossier/20014/7/11/6 as Used in Cosmetics, https://www.cir-safety.org/sites/default/files/RR_Sweet%20Almond.pdf
POD (NOAEL)	N/A

INCI	Carbomer
Presence in the annexes	None
Description	Carbomer is manufactured by reflux polymerization of acrylic acid in an inert solvent in the presence of a catalyst. In doing this, a closed system, free of oxygen and water, is used. It is a large molecule and remains in particulate form (dispersed) in final preparations and thus will not likely cross the stratum corneum to induce systemic toxicity.
Oral toxicokinetics	N/A
Inhalation toxicokinetics	N/A
Dermal absorption	Does not penetrate
Skin irritation	low potential for irritation
Eye irritation	zero to moderate
Skin sensitisation	low potential for sensitisation
Acute toxicity	Acute oral studies with rats, guinea pigs, mice, and dogs showed that Carbomer has low toxicity when ingested. The inhalation LC50 of Carbomer in albino rats was 1.71 mg/l. The dermal LD50 of rats exposed to Carbomer was > 3 g/kg.
Repeated dose studies	Feeding of rats with doses up to 5.0 g/kg/day Carbomer (49 days) and of rats and dogs with up to 5.0% Carbomer in the diet (21 days) resulted in lower than normal body weights. In rats fed Carbomer at dietary levels of 5.0% for 90 days, absolute liver weights and liver to body and brain weight ratios were reduced, but no pathological changes were observed. Rats fed Carbomer at dietary levels of 0.1%, 0.5%, or 5.0% for 6.5 months exhibited various organ weight changes. In dogs fed 0.5 or 1.0 g/kg/day Carbomer for 6.5 months, gastrointestinal irritation and marked pigment deposition within Kupffer cells of the liver were observed. In another study, there were no significant effects in dogs fed up to 1.0 g/kg/day Carbomer for 32 months.
Mutagenicity/genotoxicity	N/A
Carcinogenicity	N/A
Reproductive toxicity	N/A
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. CIR Expert Panel Amended Safety Assessment of Acrylates Copolymers as Used in Cosmetics

POD (NOAEL)	N/A
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INCI	Panthenol
Presence in the annexes	0
Description	Precursor of vitamin B5 (pantothenic acid), provitamin B5. Panthenol after penetration into the skin is converted into pantothenic acid. Both compounds have the same biological activities. Vitamin B is found in every living cell.
Oral toxicokinetics	With a molecular weight of 205.25 g/mol, a logKow value of -1.06 and a water solubility of >509 g/L D- Panthenol is likely to be absorbed in the GI tract. extensive gastrointestinal absorption is expected for D- Panthenol based on physical chemical properties.
Inhalation toxicokinetics	Based on the low vapour pressure of 0.000056 Pa D- Panthenol does not vaporise in a sufficient manner to become available for inhalation exposure. Exposure to aerosols may occur. However, due to the high water solubility the substance is likely to solve in the mucus lining of the respiratory tract but subsequent systemic absorption is not likely.
Dermal absorption	The substance characteristics and physical-chemical properties indicate that dermal absorption is likely. The physical state favours quick dermal absorption with liquids taken up more readily than dry particulates.
Skin irritation	not irritating
Eye irritation	not irritating
Skin sensitisation	not a dermal sensitizer
Acute toxicity	Oral: The LD50 was > 10000 mg/kg bw. No death were observed during a 7 h-inhalation exposure of rats to a saturated atmosphere of the test item at room temperature. Dermal: oral: The LD50 was > 10000 mg/kg bw. No death were observed during a 7 h-inhalation exposure of rats to a saturated atmosphere of the test item at room temperature.
Repeated dose studies	13 weeks, rat, The subchronic toxicity following voluntary consumption of the test item by male and female rats in drinking water in the concentrations of 200, 50 and 20 mg/kg bw/day for a 90 day period showed essentially negative results. The test item was concluded to be relatively innocuous under the conditions of the experiment. Signs of a mild eosinophilia appeared to be present among the experimental animals, but this has been seriously questioned. The no observed adverse effect level (NOAEL) under the conditions of this study was considered to be 200 mg/kg bw/day. NOAEL is the highest tested dose, but no adverse effects was reported; A read across approach was performed with the supporting substance DL-Ethylpanthenol. The repeated dose toxicity of DL-Ethylpanthenol was investigated in a subacute 28-day GLP and guideline oral toxicity study by daily gavage in the rat, the dose levels for the 28-day toxicity study were selected to be 0, 50, 200 and 1000 mg/kg bw/day. From the results presented in this report a definitive No Observed Effect Level (NOEL) of 1000 mg/kg bw/day was established.
Mutagenicity/genotoxicity	D,L-Panthenol was not mutagenic in the bacterial reverse mutation test using bacteria.
Carcinogenicity	N/A
Reproductive toxicity	There were no test article-related clinical findings or effects on maternal body weight, body weight gains, or food consumption observed at any dosage level. In addition, there were no test article-related effects on embryo/fetal development at any dosage level. Based on the results of this study, a dosage level of 1000 mg/kg/day, the highest dosage level evaluated, was considered to be the no-observed-adverse-effect level (NOAEL) for maternal toxicity and

	embryo/fetal development when DL- Ethyl Panthenol was administered orally by gavage to bred Crl:CD(SD) rats.
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. ECHA https://echa.europa.eu/sl/registration-dossier/-/registered-dossier/14227/7/9/3
POD (NOAEL)	1000 mg&kg bw/day from 28 days study

INCI	Urea
Presence in the annexes	None
Description	Urea is produced in the body (liver) of mammals as a consequence of normal physiological processes, primarily by the detoxification of ammonia resulting from protein catabolism, via the urea cycle.
Oral toxicokinetics	The molecular weight below 500g/mol, the absence of ionisable groups, the hydrophilic properties and the particle size, are in favor of oral absorption. When SC (subcutaneous) versus oral LD50 values were compared, a factor of 0.57-0.63 was reached. Based on these data, an oral absorption of 60% is considered
Inhalation toxicokinetics	Based upon the solid state with high particle size (not reaching the thoracic airways), low vapor pressure and high water solubility, deposition in the lower airways is assumed most unlikely and absorption by inhalation is considered to be negligible. From a conservative viewpoint, a 10 % inhalation absorption is considered.
Dermal absorption	Based upon the low molecular weight, solid state and low vapor pressure (not evaporating), and the fact that the structure is not binding to skin molecules, dermal absorption is possible. However, the high water solubility, log P <0 and absence of irritation are not in favour of dermal absorption. Experimental data on the absorption of urea across normal human skin in vitro is reported to be up to 9.5%. From a realistic conservative viewpoint, a 10 % dermal absorption is considered.
Skin irritation	not irritating
Eye irritation	not irritating
Skin sensitisation	not sensitising
Acute toxicity	LD 50 14 300 mg/kg bw
Repeated dose studies	In 12 -month carcinogenicity screening assays (Fleischman et al, 1980), F-344 rats and C57BL/6 mice (50/sex/group) were exposed to urea in the diet at concentrations of 4500, 9000 or 45000 ppm for 12 months. There were no signs of toxicity. Survival and bodyweights were unaffected by treatment. Gross and microscopic pathology did not reveal any treatment-related effects. It is concluded that urea is of very low chronic toxicity. Using default conversion factors, the dose level of 45000 ppm is calculated to be equivalent to approximately 2250 mg/kg bw/d in the rat and 6750 mg/kg bw/d in the mouse.
Mutagenicity/genotoxicity	No classification is proposed for genotoxicity. Urea is produced by the body in large quantities as a normal product of metabolism and is present in the bloodstream at high concentrations. Urea is therefore considered extremely unlikely to be genotoxic.
Carcinogenicity	No classification is proposed for carcinogenicity. There is no evidence from animal studies that urea is carcinogenic. The physiological role of urea and level of production by the human body indicates that the substance is not carcinogenic.
Reproductive toxicity	No standard studies are available. It is considered extremely unlikely that occupational, primary or secondary exposure to urea will result in any effects

	on fertility as the levels of exposure will be insignificant compared to those present in the body as a result of protein catabolism.
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. ECHA https://echa.europa.eu/registration-dossier/-/registered-dossier/16152/7/9/1
POD (NOAEL)	2250mg/kg bw/day, 12 months, rat

INCI	Triethanolamine
Presence in the annexes	None
Description	TEA is an amino alcohol commercially produced by aminating ethylene oxide with ammonia. CIR Expert Panel: Safe for use in cosmetic formulations designed for discontinuous, brief use followed by thorough rinsing from the surface of the skin; in products intended for prolonged contact with the skin, the concentration of diethanolamine should not exceed 5%; should not be used with products containing N-nitrosating agents. The TEA contains small amounts of diethanolamine and ethanolamine. The TEA is reactive and bifunctional, combining the properties of alcohols and amines. The reaction of ethanolamines and sulfuric acid produces sulfates. The TEA can act as an antioxidant against the autoxidation of fats of both animal and vegetable origin. Of concern in cosmetics is the conversion (N-nitrosation) of secondary amines (R1-NH-R2), such as diethanolamine (wherein R1 and R2 are each ethanol), into N-nitrosamines that may be carcinogenic. Tertiary alkyl amines (NR1R2R3), such as TEA (wherein R1, R2, and R3 are each ethanol), however, do not tend to react with N-nitrosating agents to directly form nitrosamines. However, tertiary amines can act as precursors in nitrosamine formation by undergoing nitrosative cleavage (eg, 1 ethanol functional group can be cleaved off from TEA to generate diethanolamine). The resultant secondary amine (ie, diethanolamine) can then be N-nitrosated (ie, to N-nitrosodiethanolamine [NDELA]). Accordingly, TEA can react, in a formulation or in vivo, with nitrites or oxides of nitrogen to form a nitrosamine. Nitrous anhydride is the oxide of nitrogen that most commonly initiates nitrosation in vivo.
Oral toxicokinetics	N/A
Inhalation toxicokinetics	Coupled with the small actual exposure in the breathing zone and the concentrations at which the ingredients are used, this information suggested that inhalation would not be a significant route of exposure that might lead to local respiratory or systemic toxic effects.
Dermal absorption	Studies in experimental animals indicated that TEA is absorbed through the skin. In mice, most of the topically applied ¹⁴ C-TEA is absorbed, and only 2% to 11% is detected at the site of application after 48 hours (Dow 1988,1989; Stott, 2000). The dermal absorption of TEA in rats was less extensive and much slower than in mice (Dow, 1988,1989).

Skin irritation	not irritating
Eye irritation	not irritating
Skin sensitisation	not sensitising
Acute toxicity	Acute toxicity data indicate low toxicity: in rats the oral LD50 was 6400 mg/kg bw; in rabbits the dermal LD50 was > 2000 mg/kg bw. Inhalation exposure for 8 hours to vapour saturated with TEA failed to cause any deaths in rats (LC50 was not determined).
Repeated dose studies	Oral: In a sub-chronic oral toxicity study, a NOAEL of 1000 mg/kg bw/day was established, the highest dose tested. Dermal (90 days): In a sub-chronic dermal toxicity study, NOAELs of 125 and 250 mg/kg bw/day were established for local effects for males and females. Systemic NOAELs of 125 and 250 mg/kg bw/day were determined for males and females, respectively, based on kidney effects.
Mutagenicity/genotoxicity	TEA did not cause gene mutations in <i>Salmonella typhimurium</i> and <i>Escherichia coli</i> (Ames test), nor were chromosomal aberrations or sister chromatid exchanges induced in Chinese hamster ovary cells observed. An in vitro gene mutation assay (mouse lymphoma (L5178Y TK+/-) forward gene mutation assay) was also negative. All tests were performed in the absence and presence of metabolic activation.
Carcinogenicity	An increase in tumor incidence was neither observed in rats treated dermally or orally with TEA for 2 years, nor in mice exposed to TEA by oral administration for 2 years. In a dermal mouse carcinogenicity study there was an increased incidence of liver tumors in females at all doses tested. Mechanistic research specifically on TEA indicates that, to the extent TEA can potentially induce tumors in mice, it does so by a mechanism that is not relevant to humans. Therefore, based on the available data, TEA is not considered carcinogenic for humans.
Reproductive toxicity	No two-generation reproduction toxicity study is available for TEA. Under the conditions of a two-generation reproduction toxicity study with MEA HCl, the NOAEL for systemic toxicity and fertility, reproductive performance in parental F0 and F1 Wistar rats is 300 mg/kg bw/day. The NOAEL for pre-and postnatal developmental toxicity in their offspring is 1000 mg/kg bw/day.
Photosensitivity	TEA at cosmetic use concentrations was not phototoxic or photoallergenic in non-human and human studies, and TEA-lauryl sulfate was not photoallergenic in human studies.
Endocrine effects	N/A
Reference	1. ECHA https://echa.europa.eu/registration-dossier/-/registered-dossier/15134/7/9/1 2. CIR Expert Panel Safety Assessment of Triethanolamine and Triethanolamine-Containing Ingredients as Used in Cosmetics
POD (NOAEL)	300 mg/kg bw/day, repr. Toxic. 2 generations, oral unknown

INCI	Achillea Millefolium Extract
Presence in the annexes	None
Description	Achillea millefolium is an herbaceous plant with characteristic narrow, oblong, multiple pinnate leaves. A sample of an achillea millefolium extract (aqueous) mixture (water 73.5%, butylene glycol 20%, pentylene glycol 5%, achillea millefolium extract 1%, xanthan gum 0.5%) contained 3.37% polyphenols, 61.25% proteins, and 38.12% sugars. Achillea millefolium is reported to contain linalool (1-4,000 ppm), thujone, quercetin, a-peroxyachifolid, and hydroquinone. The CIR Panel concluded that achillea millefolium extract,

	achillea millefolium flower extract (Note 1), and achillea millefolium flower/leaf/stem extract (Note 1) are safe in cosmetics when formulated to be nonsensitizing.
Oral toxicokinetics	No new toxicokinetics data were identified or made available for review. However, since botanical extracts are mixtures, toxicokinetic data would only have meaning for individual constituents of the extract, not for the extract as a whole.
Inhalation toxicokinetics	There were no inhalation toxicity data available. However, the Panel believes that the sizes of a substantial majority of the particles of the products containing these ingredients, as manufactured, are larger than the respirable range and/or aggregate and agglomerate to form much larger particles in formulation.
Dermal absorption	N/A
Skin irritation	not irritating
Eye irritation	not irritating
Skin sensitisation	Achillea millefolium is a member of the Asteraceae (formerly Compositae) family, which is known to be sensitising. An aqueous A millefolium extract was not a sensitizer in an LLNA at 1%. Two products containing Achillea millefolium extract up to 0.001133% were not sensitizing in HRIPTs. A product containing achillea millefolium extract at 0.04% was neither irritating nor sensitizing. A woman was reported to develop an allergic reaction to A millefolium after working with dried flowers.
Acute toxicity	Acute dermal toxicity data on A millefolium (yarrow)-derived ingredients were not found in the published literature, and no unpublished data were submitted. The oral LD50 of the mixture containing an aqueous extract of A millefolium was reported to be >2,000 mg/kg in female rats.
Repeated dose studies	An aqueous A millefolium extract (0.3, 0.6, 1.2 g/kg/d; leaves, stalks, stems) orally administered to male and female Wistar rats (n ¼ 10/sex) for 28 or 90 consecutive days produced no signs of toxicity. The authors concluded that the rats had no treatment-related toxicological or histopathological abnormalities. An ethanol (60%) multiherb mixture (20 mg/kg/d) that included achillea millefolium extract (3.5%; 0.7 mg/kg), orally administered to CBA/HZb mice (n ¼ 6) for up to 6 months, caused no clinical signs.
Mutagenicity/genotoxicity	In an Ames test using Salmonella typhimurium (TA98, TA100, TA102, TA1535, TA1537), the mixture containing an aqueous extract of A millefolium (0.06-5 mL) was not mutagenic with or without metabolic activation. In 2 micronucleus tests using V79 cells, the mixture containing an aqueous extract of A millefolium (up to 15 000 mg/mL) was not clastogenic or aneugenic with or without metabolic activation. 6 In a gene mutation assay using mouse lymphoma L5178Y TKþ/, a product (up to 5 mL/mL) that contained an aqueous extract of A millefolium (0.5%) was not mutagenic with or without metabolic activation.
Carcinogenicity	N/A
Reproductive toxicity	An ethanol (45%) A millefolium extract (2.8 g/kg/d; 56 times the equivalent of a human dose of 50 mg/kg/d) was not maternotoxic when orally administered to Sprague Dawley rats (n ¼ 5) but caused decreased body weights in fetuses. An aqueous A millefolium leaf extract (0.3, 0.6, 1.2 g/kg/d) orally administered to male Wistar rats (n ¼ 10) for 90 days was not toxic nor caused any clinical or behavioral signs, but there was an increase in abnormal sperm in the males in the high-dose group. An ethanolic A millefolium flower extract (200, 400,

	800 mg/kg) IP or orally administered to male albino Wistar rats (n ¼ 5) every other day for 22 days caused no changes in the low-dose IP group and the low- and mid-dose oral groups; however, there were abnormalities in the development of sperm in the mid- and high-dose groups. Although A millefolium-derived ingredients caused an increase in abnormal sperm and damage to male organs in rats, these effects were observed at levels much greater than any from exposure to cosmetics.
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. CIR Expert Panel Safety Assessment of Achillea millefolium as Used in Cosmetic, https://online.personalcarecouncil.org/ctfa-static/online/lists/cir-pdfs/PRS627.pdf
POD (NOAEL)	400mg/kg bw/day from reprod. Toxicity test 20 days

INCI	Aesculus Hippocastanum Extract
Prilozi Pravilnika o kozm.proizvodima	None
Opis supstance	Ekstrakt Aesculus Hippocastanum je ekstrakt divljeg kestena, Aesculus hippocastanum L., Hippocastanaceae. Ekstrakt se proizvodi od biljnog leka odgovarajućim postupkom korišćenjem hidroalkoholnog rastvarača koji je po jačini ekvivalentan etanolu. Seme A. hippocastanum L. sadrži 3-10% mešavine aciliranih triterpenskih glikozida (saponina). Oni su zasnovani na dva aglikona – protoaescigeninu i baringtogenolu C, koji se razlikuju samo na C-24, koji je hidroksilisan u protoaescigeninu. Dobro utvrđeni suvi ekstrakt treba da bude standardizovan na približno 2,4 puta manji sadržaj triterpenskih glikozida, odnosno 21 mg izražen kao protoaescigenin umesto prethodnog sadržaja od 50 mg izraženog kao escin (50 mg/2,4=20,8 mg≈21 mg). Isti pristup se primenjuje i na suvi ekstrakt koji odgovara određenoj količini triterpenskih glikozida, rastvarač za ekstrakciju etanol 25-50% (V/V). Rezultati relevantnih eksperimentalnih studija u prilog predloženim tradicionalnim indikacijama za upotrebu na koži su ograničeni.
Toksikokinetika oral	Neke nekliničke farmakokinetičke studije su sprovedene na escinu, ali nisu predstavljeni ubedljivi dokazi da se farmakološka aktivnost ekstrakta semena divljeg kestena može pripisati escinu. Trenutno su farmakokinetički podaci o apsorpciji, distribuciji i eliminaciji escina od ograničenog značaja. Nema indikacija da se iz escina formiraju toksični metaboliti. Nedostaje klinička potvrda o dejstvu escina na CIP-enzime. Bioraspoloživost oralno primenjenog escina je samo oko 1,5% date doze.
Toksikokinetika inhalacija	N/A
Dermalna apsorpcija	Alkohol: dermalna apsorpcija iz in vitro studija: 21% maksimalne apsorpcije (gori slučaj) Apsorpcija u realnim uslovima tipično 1-2%.
Iritacija kože	N/A
Iritacija oka	N/A
Senzitizacija kože	N/A
Akutna toksičnost	LD50-vrednosti (mg/kg t.m.) za ekstrakt semena divljeg kestena (suvi ekstrakt, DER 5:1, rastvarač za ekstrakciju 50% V/V, standardizovan za sadržaj od 50 mg escina u 240-290 mg ekstrakta) za pacov je 2150 mg/kg tm/dan za oralnu upotrebu. Za aescin su dostupni sledeći podaci u Blaschek et al., 1992: LD50 (mg/kg t.m.) intravenozno: Miš 9.3, zec 5.0, pacov 16.8, zamorac 9.1, svinja 4, pas 3. Intraperitonealnom primenom LD50 od 17 mg/kg je određena za pacove.
Studije ponovljenih doza	Dnevno i.v. doze ekstrakta semena divljeg kestena (suvi ekstrakt, DER 5:1, rastvarač za ekstrakciju 50% V/V, standardizovan za sadržaj od 50 mg escina u

	240-290 mg ekstrakta) na 9, 30 ili 90 mg/kg t. davana pacovima tokom 60 dana. Smatralo se da je nivo doze bez efekta oko 30 mg/kg t. Ni toksični efekti, ni oštećenje organa nisu primećeni nakon 34 nedelje oralne primene ekstrakta semena divljeg kestena (suvi ekstrakt, DER 5:1, rastvarač za ekstrakciju etanol 50% V/V, standardizovan za sadržaj 50 mg escina u 240-290 mg ekstrakta) za pse (2 mužjaka, 2 ženke po dozi i kontrolnu grupu) u dozama od 20, 40 ili 80 mg/kg telesne težine dnevno (5 dana u nedelji) i za pacove (20 mužjaka, 20 žena po dozi i kontrolnoj grupi) u dozama od 100, 200 i 400 mg/kg tm dnevno. Najviša doza korišćena kod pasa odgovarala je 8 puta većoj od uobičajene terapijske doze kod ljudi. Najveća doza proučavana kod pacova iznosi 40 puta veću od terapijske doze za ljude.
Mutagenost/genotoksičnost	U Ames testu mutagenosti, koristeći Salmonella tipimurium soj TA 98, komercijalni suvi ekstrakt semena (nema dodatnih informacija o ekstraktu dostupnim u referenci) dao je negativan odgovor bez aktivacije, ali sedmični pozitivan odgovor (faktor 2-3) sa S9 aktivacija. Tečni ekstrakti semena divljeg kestena dali su slabo pozitivan odgovor (faktor 2-3) bez aktivacije i negativan sa aktivacijom. Autori su sugerisali da je kvercetin možda glavni mutageni princip u ovim ekstraktima
Karcinogenost	N/A
Reproduktivna toksičnost	Mladi pacovi su tretirani 2 puta 5 mg/kg aescina u dobi od 32 dana. Nakon što su dostigle plodnost, pregledani su bubrezi, testisi i spermatozoidi. Visoka doza korišćenog escina nije uticala na plodnost i nije mogla biti otkrivena nefrotoksična aktivnost. Nakon dnevne oralne primene ekstrakta semena divljeg kestena (suvi ekstrakt, DER 5:1, ekstrakcija rastvarač etanol 50% V/V, standardizovan za sadržaj od 50 mg escina u 240-290 mg ekstrakta) za pacove i zečevi u dozama od 100 i 300 mg/kg t. primećeno u studijama teratogenosti. Kod 300 mg/kg tm za zečeve, značajno smanjenje ($p < 0,001$) u primećena je srednja težina fetusa. 300 mg/kg telesne težine je približno 30 puta više od preporučene terapijske doze za ljude.
Fotosenzitivnost	N/A
Endokrina aktivnost	N/A
Reference	1. EMA Assessment report on Aesculus hippocastanum L., semen , https://www.ema.europa.eu/en/documents/herbal-report/assessment-report-aesculus-hippocastanum-l-semen-final-revision-1_en.pdf
Odabrani POD (NOAEL)	30 mg/kg bw/day i.v. 60 dana, pacov

INCI	Capsella Bursa-Pastoris Extract
Prilozi Pravilnika o kozm.proizvodima	None
Opis supstance	Tečni ekstrakt (DER 1:1), rastvarač za ekstrakciju etanol 25% V/V. Capsella bursa-pastoris je uobičajena biljka koja raste do 40 cm i autohtono je u Evropi, zapadnoj Africi i Aziji. Sastojci: Flavonoidi (a.o. flavonglikozidi): kvercetin, tricin, diosmetin, kempferol, luteolin, hesperitin i derivati glikozida (npr. rutin, diosmin, hesperidin, luteolin-7-rutinozid, luteolin-7- galaktozid, kvercetin-3-rutinozid). Amini: holin (1%), acetilholin, histamin, aminokiseline (22, a.o. prolin, tiramin i ornitin), (poli)peptidi (a.o. a- i gaminobutirna kiselina, a-aminoadipinska kiselina) i proteini, Alifatične i fenolne kiseline: hlorogenska, vanilna, siringinska, fumarna kiselina, isparljivo ulje, sa najmanje 74 identifikovane komponente, sa kamforom kao glavnim konstituentom (0,02%), smolom, saponini. Na osnovu marketinškog iskustva, podaci dobijeni od država članica EU nikakve mere farmakovigilance nisu

	zabeležene nakon upotrebe preparata Capsella (na osnovu podaci dobijeni od 12 država članica, stanje 17. avgusta 2010.).
Toksikokinetika oral	Metabolizam sinigrina u ljudskom telu je složeniji. U dinamičkom in vitro modelu debelog creva ispitivana je proizvodnja alil izotiocijanata iz sinigrina nakon inokulacije kompleksnom mikroflorom ljudskog porekla. Kako je samo mali deo (srednjih 19%) sinigrina pretvoren u alil izotiocijanat, sugerisano je da se alil izotiocijanat dalje pretvara u druge, još nepoznate, metabolite (Krul et al., 2002). Izotiocijanat se može dalje razgraditi da bi se dobio tiocijanat jon. Ova komponenta može izazvati struma kod ljudi kada je jod u ishrani nedovoljan. Međutim, nema dokaza koji podržavaju uzročnu ulogu glukozinolata u ishrani u ljudskoj strumi (NZFSA, 2010).
Toksikokinetika inhalacija	N/A
Dermalna apsorpcija	N/A
Iritacija kože	moгуći iritans
Iritacija oka	N/A
Senzitizacija kože	N/A
Akutna toksičnost	LD50 1.5 g/kg bw (miš, inj.) i 31.5 g/kg bw (mice, subcutaneous injection)
Studije ponovljenih doza	N/A
Mutagenost/genotoksičnost	N/A
Karcinogenost	Međunarodna agencija za istraživanje raka (IARC) izveštava da nema adekvatnih dokaza o kancerogenosti alil izotiocijanata kod ljudi i ograničenih dokaza o kancerogenosti alil izotiocijanata kod eksperimentalnih životinja
Reproduktivna toksičnost	Upotreba Capsella se ne preporučuje tokom trudnoće i dojenja, jer nedostaju podaci o reproduktivnoj toksičnosti. Podaci o plodnosti nisu dostupni.
Fotosenzitivnost	N/A
Endokrina aktivnost	N/A
Reference	1. European Medicines Agency, Assessment report on Capsella bursa-pastoris (L.) Medikus, herba, https://www.ema.europa.eu/en/documents/herbal-report/final-assessment-report-capsella-bursa-pastoris-l-medikus-herba-first-version_en.pdf
Odabrani POD (NOEL)	N/A

INCI	Calendula Officinalis Flower Extract
Prilozi Pravilnika o kozm.proizvodima	None
Opis supstance	CTFA udruženje: specifikacija za neven od Alban Muller International-a koja je uključivala 12% glikozida (otprilike 6% sluzi), 5,3% lipida, 10% minerala i 0,02% eteričnog ulja sa oksigeniranim seskviterpenskim derivatima, zajedno sa organskim kiselinama; fenolna jedinjenja, uključujući fenolne kiseline (salicilna kiselina), flavonoide (flavonol heterozidi izorhamnetina i kvercetina i izorhamnetin 3-rutinozid) i tanine; terpenoide, uključujući triterpene (saponine; npr., bidesmozide i monodezmozide oleanolne kiseline i fitosterola i taraksasterola), karotenoide (karotene, likopene, violaksantine, flavoksantine) i a- i b-amirine, arnidiolradiol, hendiol i alula; lignin). Hidroetanolni ekstrakti pripremljeni maceracijom i perkolacijom će sadržati srednje polarne i polarne klase kao što su flavonoidi, terpenoidni glikozidi, karotenoidi, kumarini, fenoli, kiseline i tanini. Superkritični CO2 ekstrakti će sadržati jedinjenja niskog polariteta, sa nekim srednjim polaritetima, ali će specifična jedinjenja zavisiti od pritiska i ko-rastvarača, kao što je etanol. Lipofilne ekstrakcije (npr. biljnim ili mineralnim uljem, oktil palmitatom) će uključivati lipofilne ugljovodonike; parafini; masne kiseline i estri masnih kiselina; steroidi; tocopherols; apolarni karotenoidi; mono-, seskvi- i triterpenoidni estri; i triterpenski mono-alkoholi

	i dioli. Komitet eksperata za kozmetičke proizvode Saveta Evrope objavio je monografiju o kvercetinu, napominjući da se ova hemikalija može naći u mnogim biljkama, uključujući C officinalis. Kvercetin, na procenjenim nivoima u ishrani kao dodatak ishrani (200-1200 mg/d) ne izaziva štetne posledice po zdravlje. Analiza praga toksikološke zabrinutosti, koristeći konzervativne pretpostavke o dermalnoj apsorpciji, primenjena na brojne hemikalije pronađene u cveću C officinalis, rezultirala je sistemskim vrednostima izloženosti ispod praga toksikološke zabrinutosti vrednosti utvrđene za relevantnu Cramer klasu, osim za kvercetin, kempferol i izorhamnetin. Dostupni NOAEL sugeriše da ove hemikalije ne bi predstavljale nikakvu toksičnost u koncentracijama koje se koriste u kozmetici.
Toksikokinetika oral	N/A
Toksikokinetika inhalacija	N/A
Dermalna apsorpcija	N/A
Iritacija kože	nije iritans do slab iritans
Iritacija oka	nije iritans do slab iritans
Senzitizacija kože	Nije senzibilizator
Akutna toksičnost	LD50 4.64 g/kg.
Studije ponovljenih doza	Harikumar i saradnici su izvestili o 13-nedeljnoj studiji luteina i luteinskog estra izolovanih iz cvetova nevena koristeći mužjake i ženke pacova Vistar. Životinje (5 mužjaka i 5 ženki po grupi) primale su samo suncokretovo ulje (kontrola - nema detektabilnog luteina), ili 4, 40 ili 400 mg/kg luteina ili 4, 40 ili 400 mg/kg luteinskog estra putem oralne sonde. Autori su zaključili da su lutein i estri luteina nisu toksični za pacove. Kvercetin NOAEL 400 mg/kg/dan
Mutagenost/genotoksičnost	Ramos i saradnici proučavali su genotoksičnost ekstrakta C officinalis L koristeći 3 kratkoročna testa: Amesov test u S tiphimurium TA1535, TA1537, TA98 i TA100, sa i bez metaboličke aktivacije; test mitotske segregacije u diploidnom soju (D-30) Aspergillus nidulans; i test mikronukleusa na mišu. Autori su izvestili da nije bilo povećanja mutacija u bilo kom od Ames testova, sa ili bez metaboličkih aktivacija, za bilo koju koncentraciju ekstrakta. U svakom slučaju, pozitivne kontrole su dale očekivane rezultate. Autori su prijavili da nema povećanja mikronukleusa u polihromatskim ćelijama i nema promene u odnosu polihromatsko-normohromatskih ćelija.
Karcinogenost	Međunarodna agencija za istraživanje raka (IARC) je procenila potencijal kancerogenosti kumarina kod ljudi, zaključivši da se ova hemikalija ne može klasifikovati u pogledu njene kancerogenosti za ljude. U proceni je navedeno da postoje ograničeni dokazi na eksperimentalnim životinjama o kancerogenosti kumarina. . Konkretno, kod miševa od 1 soja, proizveo je povećanje adenoma i karcinoma pluća i kod mužjaka i kod ženki i kod hepatocelularnih adenoma kod ženki. Nije bilo povećanja incidencije tumora kod drugog soja miševa. U jednoj studiji na pacovima, kumarin je proizveo nisku incidencu adenoma bubrežnih tubula kod mužjaka, što se vidi tek nakon koraka sečenja bubrega. Hard i saradnici50 su ponovo procenili tumore bubrega i histopatologiju bubrega u dvogodišnjoj studiji kancerogenosti pacova (Nacionalni toksikološki program [NTP]) kvercetina kod mužjaka i ženki pacova dati 0, 1000, 10 000 i 40 000 ppm dnevno. Kvercetin, u ovom modelu, pogoršava hronična nefropatija, koja dovodi do tumora bubrega i smatra se sekundarnim mehanizmom za razvoj tumora bubrega. Autori su takođe predložili da pojedinačni adenokarcinom, zajedno sa grupom od 4 druge lezije (u rasponu od hiperplazije do adenoma) imali su jedinstveni fenotip povezan sa neoplazmama spontanog i porodičnog porekla. Pregledom dvogodišnje NTP studije kancerogenosti predstavljene

	ranije u ovom dokumentu, autori su zaključili da je težina dokazi podržavaju nalaz da kvercetin, na procenjenim nivoima u ishrani kao dodatak ishrani (200-1200 mg/d) ne izazivaju štetne posledice po zdravlje
Reproduktivna toksičnost	28-dnevna studija subhronične toksičnosti na pacovima. Korijander je davan oralno u dozama od 160, 400 i 1000 mg/kg po danu. Prijavljeno je povećanje težine jetre, degenerativne lezije u korteksu bubrega i visoka incidencija blage periportalne hepatocelularne citoplazmatske vakuolizacije. Nivo bez uočenog efekta (NOEL) je utvrđen na 160 mg/kg po danu. Pregledom je zaključeno da su se štetni efekti na reprodukciju i razvoj javili na nivoima toksičnih za majku i ne na nivoima koji nisu bili toksični za majku.
Fotosenzitivnost	Not phototoxic
Endokrina aktivnost	N/A
Reference	1. ECHA https://echa.europa.eu/registration-dossier/-/registered-dossier/21622/4/8 , 2. CIR Expert Panel Final Report of the Cosmetic Ingredient Review Expert Panel Amended Safety Assessment of Calendula officinalis-Derived Cosmetic Ingredients, https://online.personalcarecouncil.org/ctfa-static/online/lists/cir-pdfs/PRS514.pdf
Odabrani POD (NOAEL)	400 mg/kg bw/day, 13 nedelja, pacov

INCI	Geranium Robertianum Extract
Prilozi Pravilnika o kozm.proizvodima	None
Opis supstance	Geranium robertianum L., poznatiji kao Herb Robert ili Crveni Robin, je spontana, zeljasta biljka koja se dugo koristi u narodnoj medicini nekoliko zemalja. U hemiji G. robertianum jasno dominiraju fenolni sastojci, a najviše proučavane klase jedinjenja su tanini, flavonoidi i fenolne kiseline. Utvrđeno je da je sadržaj elagitanina u vodeno-metanolnom ekstraktu G. robertianum iz Kembridža (Engleska), izražen kao heksahidroksidifenilglukoza (HHDPG), 5% (HHDPG/suva težina). Geraniin je identifikovan u ekstraktima acetona/vode iz listova Geranium robertianum poreklom iz Japana na čak 9,8%. Rod geranijuma pokazuje širok spektar flavonoidnog uzorka, kvercetin i kempferol su prisutni u skoro svakoj vrsti.
Toksikokinetika oral	N/A
Toksikokinetika inhalacija	N/A
Dermalna apsorpcija	N/A
Iritacija kože	N/A
Iritacija oka	N/A
Senzitizacija kože	N/A
Akutna toksičnost	LD50 2000 mg/kg b. w.
Studije ponovljenih doza	Kvercetin NOAEL 400 mg/kg b. w./dan. Nivo geranina bez uočenih štetnih efekata je ispod 2000 mg/kg, dok je nivo ekstrakta obogaćenog geraninom do 2000 mg/kg.
Mutagenost/genotoksičnost	N/A
Karcinogenost	N/A
Reproduktivna toksičnost	N/A
Fotosenzitivnost	N/A
Endokrina aktivnost	N/A
Reference	1.V. C. Graça,a,b, Isabel C.F.R. Ferreirab, P. F. Santos, Phytochemical composition and biological activities of Geranium robertianum L.: a review 2. Mohanambal Moorthy,a Joon Joon

	Khoo, and Uma D. Palanisamy, acute oral toxicity of the ellagitannin geraniin and a geraniin-enriched extract from <i>Nephelium lappaceum</i> L rind in Sprague Dawley rats
Odabrani POD (NOAEL)	Quercetine NOAEL 400 mg/kg bw/day.

INCI	Chamomilla Recutita Flower Extract
Prilozi Pravilnika o kozm.proizvodima	None
Opis supstance	Cvetovi <i>Chamomilla recutita</i> (<i>Matricaria</i>) sadrže isparljivo ulje (0,24%-2,0%) koje je plave boje. Dve ključne komponente (l)-a-bisabolol i chamazulen čine 50% do 65% ukupnog sadržaja isparljivog ulja. Ostale komponente ulja su: (l)-abisabolol oksid A i B, (l)-a-bisabolon oksid A, spiroetri (cis- i trans-en-indicikloetar), seskviterpeni (antektolid), kadinen, farnezen, furfural, spatulenol i proazulen (matricarin i matricin). Hamazulen se formira iz matricina tokom parne destilacije ulja. Stručna komisija CIR izrazila je zabrinutost zbog komponenti ekstrakta cveta <i>Chamomilla Recutita</i> (<i>Matricaria</i>) (tj. kvercetina i kvercetin-3-glukozida [izokvercitrina]) koje bi mogle biti genotoksičan/kancerogen.
Toksikokinetika oral	Korišćenjem in vitro membranskog difuzionog modela, većina komponenti ulja <i>Chamomilla Recutita</i> (<i>Matricaria</i>), osim chamazulena, prošla je kroz celofansku membranu.
Toksikokinetika inhalacija	N/A
Dermalna apsorpcija	N/A
Iritacija kože	nije iritans
Iritacija oka	Sedam pacijenata sa polenskom groznicom doživelo je konjuktivitis nakon ispiranja oka <i>C recutita</i> (<i>Matricaria</i>) čajem (od cveća). Rezultati provokacionog testa sa ekstraktom čaja potvrdili su da je čaj izazvao alergijski konjuktivitis. Rezultati sledećih in vitro testova sugerišu da maskara sadrži 0,01% ekstrakta <i>Chamomilla Recutita</i> (<i>Matricaria</i>) (cela biljka vodeni ekstrakt) bio je blago iritantan za oko: NRR test, REC test i HET-CAM test.
Senzitizacija kože	Senzibilizacija kože je primećena kod 18 od 24 pacijenata sa flasterom testiranim sa 1% ekstrakta <i>Chamomilla Recutita</i> (<i>Matricaria</i>) (etarski ekstrakt) i kod 4 od 5 pacijenata i 48 od 85 pacijenata sa flasterom od 2,5% <i>Chamomilla Recutita</i> (<i>Matricaria</i>) (etarski ekstrakt) u petrolatumu. Od 9 pacijenata sa pozitivnim patch test reakcijama na SL mešavinu, 5 je takođe imalo alergijsku reakciju na 2,5% <i>C recutita</i> (<i>Matricaria</i> ; delovi biljke nisu navedeni) u petrolatumu.
Akutna toksičnost	Akutna oralna toksičnost liofilizovanog vodenog ekstrakta cveća <i>C recutita</i> (<i>Matricaria</i>) je procenjena korišćenjem 2 grupe od 12 ženki miševa soja Sviss-NOS.45 Svaka grupa je primila jednu oralnu dozu od 720 ili 440 mg/kg i primećena je tokom 24 sata nakon doziranja. Nijedna životinja nije umrla i nije bilo dokaza o akutnoj toksičnosti.
Studije ponovljenih doza	Podaci o toksičnosti ponovljenih doza prikazani su u studiji o dejstvu čaja od kamilice na aktivnost jetrenih enzima faze I i II faze metabolizma kod pacova. Od osušenih cvetnih glavica <i>C recutita</i> (<i>Matricaria</i>) može se napraviti čaj od kamilice. Pet ženki Vistar pacova (starih 8-9 nedelja) imalo je slobodan pristup rastvoru čaja od kamilice (2% [vt/vol] u vodi), dok je kontrolna grupa imala pristup vodi. Posle 4 nedelje lečenja, životinje su ubijene. Gutanje rastvora čaja nije značajno uticalo na telesnu težinu, a nije bilo ni znakova grube patologije unutrašnjih organa. Odnosi težine jetre/telesne težine tretiranih pacova nisu se značajno razlikovali od kontrolnih vrednosti. Ni indukcija ni inhibicija enzima faze I ili II nisu bili statistički značajni. Kvercetin je bezbedan

	za upotrebu i kod ženki i kod mužjaka CD2F1 miševa kada se daje u dnevnim dozama ~ 12,5, 25 ili 50 mg/kg telesne mase tokom 14 nedelja (tj. 98 dana)
Mutagenost/genotoksičnost	Mešavina povezana sa ekstraktom cveta Chamomilla Recutita (Matricaria) sadrži mineralno ulje, ulje koštica prunus armeniaca (kajsije) i ekstrakt Chamomilla Recutita (Matricaria). Sadrži 1% do 4,9% ekstrakta cveta Chamomilla Recutita (Matricaria), a mineralno ulje i ulje koštica prunus armeniaca (kajsije) su rastvarači za ekstrakciju. Genotoksičnost ove smeše (u dimetil sulfoksidu [DMSO]) je procenjena u Ames testu korišćenjem sledećih sojeva bakterija sa i bez metaboličke aktivacije: Salmonella tiphimurium sojevi TA98, TA100, TA1535 i TA1537 i Escherichia coli soj VP2 uvr1010. Autori su zaključili da mešavina trgovačkog naziva nije izazvala svako biološki relevantno povećanje broja ili revertanti u bilo kom soju, sa ili bez metaboličke aktivacije.
Karcinogenost	Ekstrakt cveta Chamomilla Recutita (Matricaria) i cvetno ulje Chamomilla Recutita (Matricaria) izazvali su značajno smanjenje vitalnosti ćelija u ćelijskim linijama humanog raka.
Reproduktivna toksičnost	N/A
Fotosenzitivnost	Pet pacijenata je inicijalno testirano na patch (Finn chambers na Scanpor traci) sa 2,5% ekstrakta Chamomilla Recutita (Matricaria) (etarski ekstrakt), a rezultati su bili pozitivni kod 4 pacijenta. 5 pacijenata je takođe procenjeno standardnim fotopatch testom. Posle prvog čitanja (1. dan) usledilo je UV zračenje i drugo čitanje trećeg dana. Nisu uključeni dodatni detalji u vezi sa procedurom testiranja. Fotoagracija (ocena nije data) primećena je kod 1 od 5 pacijenata.
Endokrina aktivnost	N/A
Reference	1. CIR Expert Panel Amended Safety Assessment of Chamomilla recutita-Derived Ingredients as Used in Cosmetics 2. Patrice Cunningham, Emma Patton, Brandon N. VanderVeen, Christian Unger, Ahmed Aladhami, Reilly T. Enos, Sarah Madero, Ioulia Chatzistamou, Daping Fan, E. Angela Murphy & Kandy T. Velázquez, Sub-chronic oral toxicity screening of quercetin in mice
Odabrani POD (NOAEL)	400mg/kg bw/day, 4 nedelje, pacov, oral, konverzija sa 2%

INCI	Tocopheryl Acetate
Presence in the annexes	0
Description	Tocopheryl acetate is in terms of Regulation (EC) No 1907/2006 (REACH) exempt from the obligation to register, as sufficient information is known about this substance that it is considered to cause minimum risk because of its intrinsic properties. SCCNFP is of the opinion that alpha-tocopherol acetate does not pose a threat to the health of the consumer and therefore does not propose any restrictions or conditions on the use of alpha-tocopherol acetate in cosmetic products. The safety of Tocopheryl Acetate has also been assessed by the CIR Expert Panel. The CIR Expert Panel evaluated the scientific data and concluded that this ingredient is safe as used in cosmetics at concentrations up to 36% (100% in Vitamin E oil).
Oral toxicokinetics	In a dermal absorption study using human subjects, tocopheryl acetate was substantially absorbed in the skin, but systemic availability was not observed. Also, conversion to tocopherol was not seen. In a study using rats, approximately 6% of the applied dose penetrated into the epidermis after 5 days. Most studies found that some tocopheryl acetate was converted to tocopherol. Irradiation of the animals dosed with tocopheryl acetate resulted in a significant increase in the amount of tocopheryl acetate found in the skin

	as compared to non-irradiated animals, and irradiation significantly increased the amount of tocopherol found in the skin of test animals.
Inhalation toxicokinetics	N/A
Dermal absorption	Dermally applied tocopheryl acetate is hydrolyzed to tocopherol upon exposure to UV. In a study using rats, approximately 6% of the applied dose penetrated into the epidermis after 5 days.
Skin irritation	Not irritating (rabbit)
Eye irritation	Not irritating to moderately irritating (undiluted, rabbit)
Skin sensitisation	Not sensitizing (undiluted, human). It produced skin sensitization in one animal test, was not sensitizing in the "guinea pig maximization test", but can cause sensitization in the "open epicutaneous test" (>30% tocopheryl acetate).
Acute toxicity	Oral LD50 (rat) > 16 g/kg; Oral LD50 (mouse) > 4 g/kg
Repeated dose studies	Tocopheryl acetate was not toxic in an 8-wk feeding study. A short-term repeated dose toxicity study (28 days) resulted in the following NOAEL values (racemic tocopheryl acetate): Rat: 1111 mg/kg; Dog: 360 mg/kg.
Mutagenicity/genotoxicity	Not genotoxic in Ames tests or a chromosomal aberration assay
Carcinogenicity	Tocopheryl acetate (≤ 2 g/kg/day) was not carcinogenic in a dietary study. Dermally applied tocopheryl acetate was reported to enhance photocarcinogenesis. Orally administered tocopheryl acetate reduced the incidence of skin cancer, but toxicity was observed.
Reproductive toxicity	Oral administration of tocopheryl acetate (≤ 1.6 g/kg/day) generally did not have any reproductive or developmental effects in rabbits, hamsters, rats, or mice. Tocopheryl acetate had some effect on reducing the number of malformations observed in neonates from diabetic dams. In some studies, tocopheryl acetate potentiated the embryo-lethal effects of cortisone acetate.
Photosensitivity	Not photoallergenic (guinea pig); photoprotective effects observed in mice and humans
Endocrine effects	N/A
Reference	(1) Norwegian Food Safety Authority. Risk profile Vitamin E. 10.12.2012. (NOAEL) (2) CIR Expert Panel. Safety Assessment of Tocopherols and Tocotrienols as Used in Cosmetics. April 4, 2014. (3) Dweck A. C. Handbook of Cosmetics Ingredients - Their Use, Safety and Toxicology. Third edition, 2012.
POD (NOAEL)	360, 28days

INCI	Retinyl Palmitate
Presence in the annexes	None
Description	Vitamin A is a micronutrient essential for most mammalian species and is one of three fat soluble Vitamins. The term "Vitamin A" is used as a generic descriptor for compounds that exhibit the biological properties of retinol or other closely related naturally occurring derivatives. Vitamin A occurs in nature primarily as retinyl ester and retinol. It is an essential factor for the growth and maintenance of higher organisms. It is required for visual function, epithelial cell differentiation and reproduction.
Oral toxicokinetics	Dietary Vitamin A, which is mostly in the form of retinyl esters, is absorbed in the upper part of the small intestine by mechanisms similar to those for lipid absorption. The esters undergo hydrolysis to release retinol, which is incorporated into mixed micelles and absorbed by enterocytes, where it is bound to an intra-cellular protein called CRBP (cellular retinol binding protein II). It is then re-esterified to form retinyl esters. The esters are

	incorporated into chylomicrons and are hydrolyzed in the general circulation. Chylomicron remnants are taken up by tissues, particularly the liver. Remnants are degraded within the hepatocytes, and the released retinol is transferred to stellate cells for storage after re-esterification. In the present study on basic toxicokinetics, the exposure to retinyl palmitate and its main metabolites in plasma was examined within the framework of prenatal developmental toxicity studies in cynomolgus monkeys (Hendrickx 2000). Retinyl palmitate was administered daily to female monkeys by nasogastric intubation at dose levels of approx. 4.1, 11, 22, 44 mg/kg (7500, 20000, 40000, 80000 IU/kg) during early pregnancy (i. e. between gestational days 16 and 27). Control monkeys were administered the vehicle (physiological saline or distilled water). Administration of retinyl palmitate resulted in the formation of the following metabolites: retinol, retinyl esters, all-trans-retinoic acid, 13-cis-retinoic acid, all-trans-4-oxo-retinoic acid, 13-cis-4-oxo-retinoic acid. In a weight of evidence, the dermal absorption rate of retinol and the respective esters in humans is considered to be approx. 5% of the applied dose. Considering the comparative assessment of oral versus dermal administration observed by Nohynek et al. (Nohynek 2006), the oral absorption is expected to be extensively higher (up to 15 fold increases), which substantiates the assumption of a 100% oral absorption of retinyl palmitate.
Inhalation toxicokinetics	In light of the present data from acute oral toxicity studies, and the fact, that inhalative exposure is not considered to be the major route of exposure, an acute toxicity study via the inhalative route is scientifically not required and is not in line with animal welfare requirements.
Dermal absorption	In a weight of evidence, the dermal absorption rate of retinol and the respective esters in humans is considered to be approx. 5% of the applied dose.
Skin irritation	Although the present key study on retinyl palmitate does not meet the criteria for classification as a skin irritant according to directive 67/548/EEC, the supportive study and the findings from structurally similar substances, i.e retinyl acetate and retinyl propionate, provide evidence for dermal irritative potential of this substance group, warranting a respective classification. However, the dermal irritative potential of retinyl palmitate appears to be lower than for retinyl acetate and retinyl propionate on the basis of the present data.
Eye irritation	The present data on eye irritation do not fulfill the criteria laid down in 67/548/EEC and regulation 1272/2008/EC, and therefore, a non-classification is warranted.
Skin sensitisation	not sensitising
Acute toxicity	The acute toxicity of retinyl-palmitate in animals is low, based on the data of structural analogue retinyl-propionate. The available acute oral toxicity study upon oral application of retinyl propionate (approx. 95% purity) was selected as the key study (performed according OECD 401 guideline, BASF 10A0454/891109) The oral LD50 was > 2000 mg/kg bw in rats in this well conducted limit study. In an acute oral toxicity study in rats, available as short summary from a secondary source, a LD50 = 7910 mg/kg has been given for retinyl-palmitate (Kamm 1987).
Repeated dose studies	A repeated dose 90 days (feed) study in rats was available on structural analogue retinyl acetate that was performed similar to OECD guideline 408: The NOAEL was 1.43 -3.07 mg/kg bw/d (28 ppm) based on slight effects on a few clinical chemical parameters (increased plasma triglyceride levels, slightly increased activity of glutamate pyruvate transaminase and glutamate oxalacetate transaminase) and effects in the liver (increased absolute and relative liver weights, brightened livers, lipid accumulation of the Kupffer cells and decrease of the intrahepatocellular lipid droplets in the peripheral areas of the liver lobules). According to the dose response assessment by the

	scientific committee on food (EFSA 2006) mainly on the basis of human surveys, the lowest doses reported to produce the major adverse effects were 7500 RE/day (representing approx 13.5 mg retinyl palmitate/ day) for bulging fontanelles, hepatotoxicity and lipid metabolism, > 3000 RE/day (representing approx 5.5 mg retinyl palmitate/day) for teratogenicity. Data concerning decreased bone density with focus on postmenopausal women at lower dose levels exist, but were considered to be insufficient to provide a sound evidence of causality between vitamin A intake and increased risk for bone fractures.
Mutagenicity/genotoxicity	On the basis of all available information on genotoxicity of retinyl palmitate and relevant studies on its respective structural and metabolic analogues retinol and retinyl acetate, retinyl palmitate is defined to be non-genotoxic in a weight of evidence.
Carcinogenicity	On the basis of all available information on genotoxicity of retinyl palmitate and relevant studies on its respective structural and metabolic analogues retinol and retinyl acetate, retinyl palmitate is defined to be non-genotoxic in a weight of evidence.
Reproductive toxicity	In this array of developmental toxicity studies in cynomolgus monkeys (Macaca fascicularis) retinyl palmitate was orally (nasogastric intubation) administered at doses of 4.1, 11, 22 and 44 mg/kg bw/d (7 500, 20 000, 40 000, 80 000 IU/kg bw/d, respectively) from GD 16 to GD 27 (Hendrickx 2000). Taken together, from the variety of available data on developmental toxicity and retinoids, the cynomolgus monkey was shown to be a sensitive animal species and the results obtained are considered as most relevant for human risk assessment. In the series of prenatal developmental toxicity studies in the pregnant Cynomolgus monkeys, oral doses of retinyl palmitate at 20000 and 40000 IU/kg bw/day resulted in developmental toxicity and maternal toxicity (Hypervitaminosis A), respectively, leading to an NOAELdevelopmental of 4.1 mg/kg bw/d and a NOAELmaternal of 11 mg/kg bw/d.
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. ECHA https://echa.europa.eu/registration-dossier/-/registered-dossier/13687/7/6/2
POD (NOAEL)	1.43 mg/kg bw/day from 90 days oral study

INCI	Tocopherol
Presence in the annexes	0
Description	Tocopherol and tocopheryl acetate are generally recognized as safe (GRAS) food ingredients. Systemic exposure resulting from cosmetic use is not expected to exceed the levels found safe for food use. Together, the tocopherols and tocotrienols comprise vitamin E
Oral toxicokinetics	N/A
Inhalation toxicokinetics	N/A
Dermal absorption	The effect of vehicle on the dermal penetration of α -tocopherol was determined in micro-Yucatan pig skin. ⁴⁰ The dermal penetration of 1% tocopherol was evaluated in several vehicles, and α -tocopherol permeated into the receptor fluid regardless of the vehicle used. The greatest permeation found in viable skin + the receptor was from an emulsion vehicle that contained 10% isopropyl myristate; 12.24% of the applied dose was found in the viable skin and receptor fluid. The lowest penetration, 6.47% applied dose, was observed with an alcoholic vehicle that was 96% SD alcohol.
Skin irritation	non to mild irritating
Eye irritation	non to mild irritating

Skin sensitisation	N/A
Acute toxicity	In rats, the dermal LD50 is >3 g/kg. In mice, the oral LD50 of tocopherol is >25 ml/kg
Repeated dose studies	13 weeks study on rats, the NOAEL is set at 500 mg/kg.
Mutagenicity/genotoxicity	The test compound did not induce any relevant increase of the number of revertant colonies/plate in any of the five tester strains. Thus it can be concluded that all - rac α tocopheryl-acetate is not mutagenic in the Ames test under the described experimental conditions.
Carcinogenicity	N/A
Reproductive toxicity	No apparent differences in reproductive indices were seen between controls and treated groups of parents. Mean gestation period, litter size, sex ratio, and mortality of pups and parents were unaffected by the test substance. Hematology and clinical chemistry done after 255 days of treatment revealed no toxicologic differences between control and high dose group. None of the organ weights (neither absolute nor relative) of the treated groups were significantly different from control. Microscopic examination of tissues of high dose and control F0 and F1b animals revealed no morphological changes that were attributable to ingestion of the test substance. Ingestion of the test substance had no effect on body weight gain of the pups. Based on default values in the REACH guidance Chapter R.8: Characterisation of dose [concentration]-response for human health (Table R8 -17); using the daily intake of 20 gram/day: and the average body weight of the rats of 0.5 kg (data of the 90 day study), the NOAEL of 2% is converted to 800 mg/kg bw/day.
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. ECHA https://echa.europa.eu/sl/registration-dossier/-/registered-dossier/12703/7/9/2 2. CIR Safety Expert Panel https://www.cir-safety.org/sites/default/files/tocopherols.pdf
POD (NOAEL)	500 from 13 weeks study on rats

INCI	LAVANDULA HYBRIDA OIL
Prilozi Pravilnika o kozm.proizvodima	None
Opis supstance	Essential oil of lavender hybrida.
Toksikokinetika oral	1
Toksikokinetika inhalacija	1
Dermalna apsorpcija	1
Iritacija kože	not irritant
Iritacija oka	irritant
Senzitizacija kože	sensitizer
Akutna toksičnost	LD50>5000 mg/kg bw
Studije ponovljenih doza	In a GLP 28-day repeated dose toxicity study (similarly to OECD Guideline 407) with read-across substance coriander oil, NOAEL rats = 160 mg/kg bw/d (males and females)
Mutagenost/genotoksičnost	Ames test: non mutagenic, Micronucleus test in human lymphocytes: non clastogenic
Karcinogenost	N/A
Reproduktivna toksičnost	In a GLP reproductive and developmental toxicity screening study (similarly to OECD Guideline 421) in rats with read-across substance coriander oil, NOAEL for maternal toxicity and developmental toxicity = 500 mg/kg bw/day
Fotosenzitivnost	N/A
Endokrina aktivnost	N/A

Reference	1. ECHA https://echa.europa.eu/registration-dossier/-/registered-dossier/21156/7/9/1
Odabrani POD (NOAEL)	160mg/kg bw/day, 28 days, rat, oral

INCI	Pogostemon Cablin Leaf Oil
Prilozi Pravilnika o kozm.proizvodima	/
Opis supstance	Etarsko ulje pačulija (PEO) se dobija parnom destilacijom ili hidrodestilacijom osušenih listova Pogostemon cablina. Kao glavni sastojak PEO, pačulol je mirisni sastojak. Glavna fitokemijska jedinjenja su pačuli alkohol, a-pačulen, b-pačulen, a-bulnesen, sejšelen, norpačulenol, čestone, eugenol i čestol.
Toksikokinetika oral	0.5
Toksikokinetika inhalacija	1
Dermalna apsorpcija	0.5
Iritacija kože	nije iritans
Iritacija oka	iritans
Senzitizacija kože	Limonen, linalool, citronelol, farnesol i eugenol su sumnjivi senzibilizatori kože sa ograničenim nivoima upotrebe
Akutna toksičnost	LD 50 > 5.0 g/kg bw
Studije ponovljenih doza	Oralna (dijetalna) primena ulja pačulija pacovima u periodu od dvadeset osam uzastopnih dana pri nivoima doze do 13000 ppm rezultirala je toksikološki značajnim efektima pri 13000 ppm (ekvivalentno srednjoj postignutoj dozi od 977 mg/kg tm/dan).
Mutagenost/genotoksičnost	Ames test: nije mutagen
Karcinogenost	N/A
Reproduktivna toksičnost	NOAELparental toxicity ≥ 810 mg/kg bw/d, NOAEL reproductive and developmental toxicity ≥ 810 mg/kg bw/d
Fotosenzitivnost	N/A
Endokrina aktivnost	N/A
Reference	1. Teris A. van Beek, Daniel Joulain, The essential oil of patchouli, Pogostemon cablin: A review 2. ECHA https://echa.europa.eu/registration-dossier/-/registered-dossier/14973/7/2/1
Odabrani POD (NOAEL)	810mg/kg bw/day, oral, pacov, reprod.toks.

INCI	Melaleuca Alternifolia Leaf Oil
Prilozi Pravilnika o kozm.proizvodima	None
Opis supstance	Tea tree oil is a volatile essential oil. Tea tree oil is reported to have use as an herbal medicine; it has been used for centuries as a traditional medicine.
Toksikokinetika oral	N/A
Toksikokinetika inhalacija	100% absorbtion
Dermalna apsorpcija	The EMA monograph on Melaleuca species stated that because tea tree oil is a semi-volatile substance, the majority of an applied dose would be expected to evaporate from the skin surface before it could be absorbed into the skin. In Vitro 7.2%
Iritacija kože	slightly irritating
Iritacija oka	irritating
Senzitizacija kože	slightly sensitizing
Akutna toksičnost	LD50 was > 5 g/kg
Studije ponovljenih doza	Groups of 5 male and 5 female Sprague-Dawley rats were dosed for 28 d with tea tree oil in corn oil by gavage at doses of 0, 5, 15, and 45 mg/kg/d, in accordance with OECD TG 407.7 No mortality was observed, and no test-

	article related clinical signs of toxicity were reported. The NOAEL was determined to be 45 mg/kg/d for both male and female rats.
Mutagenost/genotoksičnost	In vitro, tea tree oil was not mutagenic in an Ames test using Salmonella typhimurium and Escherichia coli WP2 uvr A, with or without metabolic activation in chromosomal assays using Chinese hamster lung fibroblasts (V79) cells ($\leq 58.6 \mu\text{g/ml}$) or human lymphocytes ($\leq 365 \mu\text{g/ml}$), in an in vitro mammalian cell micronucleus assay using human lymphocytes ($\leq 365 \mu\text{g/ml}$), in a mammalian cell transformation assay (120 and 275 $\mu\text{g/ml}$, without and with metabolic activation, respectively), or in a Comet assay using normal human keratinocytes (HaCaT) cells ($\leq 0.064\%$). In vivo, Melaleuca Alternifolia (Tea Tree) Leaf Oil was not clastogenic in a mammalian erythrocyte micronucleus test in which mice were dosed orally with up to 1750 mg/kg bw in corn oil.
Karcinogenost	N/A
Reproduktivna toksičnost	Groups of 27 mated female Hannover Wistar rats were dosed by gavage with 0, 20, 100, and 250 mg/kg bw/d tea tree oil in PEG 400 on days 5 to 19 of gestation, in a developmental toxicity study performed in accordance with OECD TG 414. The NOAELs for maternal toxicity and for developmental toxicity (secondary to severe maternal toxicity) were 20 mg/kg bw/d tea tree oil.
Fotosenzitivnost	N/A
Endokrina aktivnost	In an opinion paper, the SCCP commented that an estrogenic potential of tea tree oil was shown in vitro, but in vivo studies were not available to elucidate the relevance of this finding. ⁶ The potentially endocrine-active constituents of tea tree oil have not been shown to penetrate the skin; therefore, the (hypothesized) correlation of gynecomastia due to the topical use of tea tree oil, in conjunction with lavender oil, in a 10-yr old male, ¹⁰² was considered implausible by the SCCP.
Reference	1. CIR Expert Panel, Safety Assessment of Melaleuca alternifolia (Tea Tree)-Derived Ingredients as Used in Cosmetics
Odabrani POD (NOAEL)	20mg/kg bw/day, short study

INCI	Eugenia Caryophyllus Leaf Oil
Prilozi Pravilnika o kozm.proizvodima	None
Opis supstance	According to the European Pharmacopoeia (1/2008: 1091), Caryophylli flos consists of the whole flower buds of Syzygium aromaticum (L.) Merrill et L.M. Perry (Syn. Eugenia caryophyllus (C. Spreng.) Bull. et Harr.) which were dried until they become reddish-brown. They contain not less than 150 ml/kg essential oil. Three main components account for nearly 99% of the essential oil: eugenol (75–88%), acetyl eugenol (4–15%), β -caryophyllene (5–14%). The medicinal use of Caryophylli floris aetheroleum can be traced in literature back to the 15th century. Clove essential oil acts in high concentrations as local irritant, allergic reactions may also be possible. However, when applied in diluted form, no reports on severe adverse events are published. When applied correctly in the proposed routes of administration, clove essential oil can be considered as clinically safe.
Toksikokinetika oral	N/A
Toksikokinetika inhalacija	N/A

Dermalna apsorpcija	N/A
Iritacija kože	In concentrated form, oil of clove may be irritating to mucosal tissues (Gruenwald et al 2004). In contrast to this Anton et al (2001) report that there is no skin irritation (undiluted oil) on hairless mice. Under occlusion the undiluted clove oil was moderately irritating in rabbits.
Iritacija oka	0
Senzitizacija kože	In a patch test study a 10% ethanol extract of Caryophylli flos was investigated among other herbal preparations used in the Traditional Chinese Medicine. Out of 30 patients 8 reacted positively to clove extract (Chen et al 2003).
Akutna toksičnost	LD50 1.8–3.72 g/kg
Studije ponovljenih doza	Clove essential oil in oral dosages of 35 or 70 mg per animal (rat) over 8 weeks was tolerated without signs of toxicity. Higher doses led to inactivity and weight loss. 105 mg/animal p.o. daily for 2 to 3 weeks led to serious liver and kidney damage and death of the animals. Clove oil is allowed as food additive and therefore an administration to food-producing animals is possible. The allowed daily intake for the United States is reported with 2.5 mg/kg
Mutagenost/genotoksičnost	The dry residue of aqueous and methanolic extracts showed mutagenic effects in the rec assay in <i>Bacillus subtilis</i> . The mutagenic activity in the Ames test on <i>Salmonella typhimurium</i> TA98 and TA100 was not assessable due to the antimicrobial action. An in vivo bone marrow micronucleus assay demonstrated that the administration of 0.5% and 2% of cloves in the diet of mice for 10 days neither significantly induced micronuclei nor could effectively modulate the 7, 12-dimethylbenz[a]anthracene genotoxicity in mice
Karcinogenost	Rats: eugenol was administered in the diet in a concentration of up to 6000 ppm. Considering the mean diet consumption per day the daily uptake of eugenol was for male rats about 1100 mg/kg body weight at the beginning to 245 mg/kg body weight at the end of the study; for female rats about 790 mg/kg body weight at the beginning to 237 mg/kg body weight at the end of the study. The study outcome was negative in both sexes.
Reproduktivna toksičnost	Domaracky et al (2007) investigated the effects of clove essential oil on the growth and development of mouse pre-implantation embryos in vivo. The animals received 0.25% essential oil in the commercial diet (= 375 mg/kg per day). Treatment with clove essential oil induced a significantly higher percentage of dead cells compared to the control group. Female CD-1 mice were given clove oil from day 6 to day 15 of gestation p. o. 2.2 to 215 mg/kg body weight daily. The foetuses were obtained on day 17. The use of clove oil had no apparent toxic effect on the implantation and survival of the mother and foetus. The number of malformations of the soft tissues and the skeletal system was not different from that in the control group spontaneously occurring.
Fotosenzitivnost	N/A
Endokrina aktivnost	N/A
Reference	1. EMA Assessment report on <i>Syzygium aromaticum</i> (L.) Merrill et L.M. Perry, flos and <i>Syzygium aromaticum</i> (L.) Merrill et L.M. Perry, floris aetheroleum
Odabrani POD (NOAEL)	2.5mg/kg bw/day ADI

INCI	Sodium Gluconate
Prilozi Pravilnika o kozm.proizvodima	None
Opis supstance	Sodium salt of gluconic acid. Listed in REACH Annex IV:exempted from registration (as well as downstream user requirements and evaluation) for all their possible uses irrespective of the tonnage in which they are manufactured or imported (currently or in the future). - GRAS as a sequestrant in animal drugs, feeds, and related products, with no limitation other than current GMP. - Inactive ingredient for approved drugs; used in oral products. Important intermediate in carbohydrate metabolism; gluconic acid is a normal metabolic product of glucose oxidation, and the amounts produced endogenously are much greater than what is consumed.
Toksikokinetika oral	N/A
Toksikokinetika inhalacija	N/A
Dermalna apsorpcija	N/A
Iritacija kože	Not irritant
Iritacija oka	Not irritant
Senzitizacija kože	N/A
Akutna toksičnost	N/A
Studije ponovljenih doza	None of the repeated dose toxicity studies of any duration (4 weeks, 6 months, or 24 months) showed any significant toxicological effects of gluconates. Potential side effects were attributed to high doses of cation intake, evidenced by results from assays designed for the gluconate anion effect specifically. The NOAEL of sodium gluconate determined from the 28 days studies on rats was equal to 1000 mg/kg bw for males and 2000 mg/kg bw for females. These compounds are neither irritant to the eye or the skin nor show sensitizing properties.
Mutagenost/genotoksičnost	Ames test, with and without metabolic activation; appropriate positive and negative controls were used. Negative result.
Karcinogenost	N/A
Reproduktivna toksičnost	no changes were observed on the reproductive organs in 28 days oral studies with sodium gluconate (dosage up to 4400 mg/kg bw) and developmental toxicity studies on GDL on different species were all negative.
Fotosenzitivnost	N/A
Endokrina aktivnost	N/A
Reference	1. CIR Ingredients Review: Safety Assessment of Monosaccharides. 2. SIDS INITIAL ASSESSMENT PROFILE, SIAM 18, 20-23 April 2004 Disaccharides, and Related Ingredients as Used in Cosmetics
Odabrani POD (NOAEL)	20mg/kg bw/day

9) Undesirable effects and serious undesirable effects

The responsible person is informed about the obligation to collect information on undesirable effects and serious undesirable effects, as well as to enter them in the Cosmetic product safety report.

10) Information on the cosmetic product

Cosmetic product DERMOTIN BIOROID contains ingredients of natural origin in which, according to data from the existing scientific literature, the presence of allergens is possible and is indicated as a challenge in assessing the safety of the products that contain them.

The product DERMOTIN BIOROID was tested for allergens presence, the results of which are listed in the attachment to this document. Based on these results together with data stated in the raw material documentation allergens that must be labelled on the packaging are LIMONENE, EUGENOL, LINALOOL.

The chemical substances and mixtures used to produce cosmetic product DERMOTIN BIOROID are widely used as ingredients for cosmetic products, they are used in concentrations that are acceptable from the point of view of safety assessment, based on which it can be concluded that the product will not cause irritation and sensitization of the skin. Eye irritation may occur in case of contact with the eyes. The product packaging clearly states the warning measure to avoid contact with the eyes.

Cosmetic product DERMOTIN BIOROID doesn't contain:

- Substances and mixtures that have been tested on animals.
- Intentionally added CMR substances
- Prohibited substances listed in the annexes of the Regulation (EC) 1223/2009
- Intentionally added nanoparticles

Part B. Cosmetic Product Safety Assessment

1) Assessment conclusion

The cosmetic product DERMOTIN BIOROID is **safe** for human health when used under normal or reasonably foreseeable conditions, in accordance with the Cosmetic Regulation (EC) 1223/2009.

2) Labelled warnings and instructions of use

Cosmetic product DERMOTIN BIOROID is intended to be used as anal region skin care cream without subsequent washing after application.

Instructions for use written on the cosmetic product:

„Apply twice a day, along with regular hygiene. The cream is of herbal origin and can be used for a long period of time without interruption.

Warnings and precautions written on the cosmetic product:

„Hypersensitivity to any of the ingredients. The product is intended for external use only. Wash hands after use. Avoid product contact with eyes. If contact occurs, wash with plenty of water. In case of extended irritation, seek medical help. “

The cosmetic product DERMOTIN BIOROID is presented in a way that clearly communicates its purpose. Precise and clearly understandable instructions are given, based on which it is considered that it cannot be misused.

The cosmetic product DERMOTIN BIOROID has characteristics in terms of appearance, odor, packaging design, appearance and content of the labelling and volume, and other characteristics of general appearance that cannot mislead the consumer about the purpose of the product and the way of its use.

3) Reasoning

Safety assessment of the cosmetic product DERMOTIN BIOROID was done considering all available relevant data, up to date regulatory requirements and the latest scientific knowledge published. Facts relevant for the safety assessment:

- To successfully assess the safety of the product, information on the body part and body surface of product application, the amount of product applied, duration and frequency of application, normal and reasonably applicable routes of exposure to the product and substances, and the target population were considered.
- All technical documentation that defines the substances and mixtures used to produce the cosmetic product, as well as the technical documentation that defines the packaging used for the packaging of the cosmetic product, is considered adequate for evaluating the product's safety.
- All information regarding the properties of the cosmetic product, its production method and the results of the conducted experimental tests submitted by the responsible person are considered adequate for evaluating the product's safety. Any further change of any parameter relevant to the product safety assessment must be considered by the responsible person and a revision of this document carried out, of which the responsible person has been informed.
- The toxicological profiles of the substances included in the composition of the cosmetic product were generated considering the latest scientific knowledge.

- All substances for which there is a legal concentration limit, and which are listed in relevant annexes of the Regulation (EC) 1223/2009 meet the legal requirements and for them the Margin of Safety (MoS) was not calculated.
- For substances for which there is no legal concentration limit and there are available data on conducted repeated dose studies, based on which it is possible to obtain the value of the No Observed Adverse Effects Level are expected (NOAEL), the Margin of Safety (MoS) was calculated, which for all given substances is greater than 100 which is considered acceptable.
- Based on the data on the potential of ingredients substances to cause skin irritation and sensitization, it was concluded that the product DERMOTIN BIOROID is not expected to cause skin irritation and sensitization. It is possible to expect that the product in case of contact with the mucous membrane of the eye will lead to slight eye irritation, which is why it is clearly indicated on the product packaging that contact with the eyes should be avoided.
- There are no expected interactions between the ingredients of the product, as well as interactions between the ingredients of the product and the packaging during the period of 24 months after production, which would lead to negative outcomes regarding the stability and efficiency of the product, which was confirmed by stability and compatibility studies.
- The method of production and packaging of the cosmetic product excludes the possibility of occurrence of microbiological contamination that would lead to endangering human health from the use of the cosmetic product, which was confirmed by the conducted stability and compatibility studies.
- Cosmetic product doesn't contain prohibited substances listed in the Annex II of the Regulation (EC) 1223/2009.
- Based on the overall appearance of the product and the content listed on the packaging, the purpose of the product as anal region skin care cream is clear. Instructions for use and warning measures are clearly indicated on the packaging.

4) Assessor's credentials and approval of part B

Safety Assessor: Vesna Milanković, FVRM Consulting, Koste Jovanovića 21, 11000 Belgrade, Serbia

Academic and expert qualifications:

- M.Sc. Chemical Engineering, 300 ECTS (Faculty of Technology, Univeristy of Novi Sad)
- Specialist Academic Studies, Cosmetology, 60 ECTS (Faculty of Pharmacy, University of Belgrade)
- Safety Assessmet of Cosmetics in the EU, 6 ECTS (Vrije University Brussel)
- Chemical Advisor License, Belgrade.

The conclusion of the safety assessment given in part B of this document for the cosmetic product DERMOTIN BIOROID is correct and the explanation provided is adequate. The assessment was done based

on the documentation submitted by the responsible person, up to the current date of updated regulatory requirements and up to the current date of published scientific findings.

Signature:



Vesna Milanković